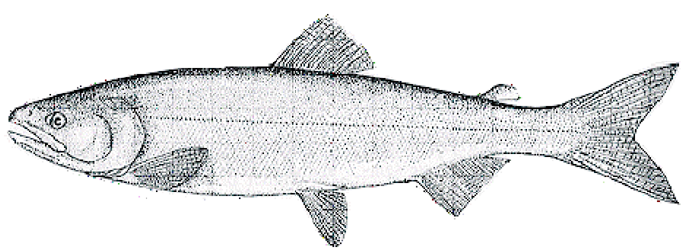
PACIFIC REGION

INTEGRATED FISHERIES MANAGEMENT PLAN

CHINOOK AND FALL CHUM SALMON

July 1, 2016 to June 30, 2017

YUKON RIVER, Y.T.



Fisheries and Oceans Canada　Pêches et Océans Canada

Canada

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DEPARTMENTAL CONTACTS

A more comprehensive list of contacts can be found online at:
http://www.pac.dfo-mpo.gc.ca/fm-gp/contacts-eng.html

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Other Key Contacts

Yukon Salmon Sub-Committee (YSSC) Office	(867) 393-6725
Pacific Salmon Commission (PSC) Office	(604) 684-8081

INDEX OF WEB-BASED INFORMATION

A more comprehensive index of web-based information is available at www.pac.dfo-mpo.gc.ca. If you don't have access to the Internet, the Fisheries and Oceans Canada office can download and print the information you need. Please allow a minimum of one week to obtain the information.

FISHERIES AND OCEANS CANADA - GENERAL INFORMATION

DFO Main Page: (www.dfo-mpo.gc.ca)
General information on the Department.

Acts, Orders, and Regulations: (www.dfo-mpo.gc.ca/communic/policy/dnload_e.htm)
Fisheries Act, Oceans Act.

Reports and Publications: (http://www.dfo-mpo.gc.ca/reports-rapports-eng.htm)
Administration and Enforcement of the Fish Habitat Protection Provisions of the Fisheries Act, Audit and Evaluation Reports - Audit and Evaluation Directorate Canadian Code of Conduct for Responsible Fishing Operations, Departmental Performance Reports, Fisheries Research Documents, Standing Committee's Reports and Government responses.

Waves: (http://www.dfo-mpo.gc.ca/libraries-bibliotheques/index-eng.htm)
DFO online library catalogue.

Pacific Salmon Treaty (http://www.psc.org/about_treaty.htm**)**
Full text of the 1999 Agreement with updates (including the *Yukon River Salmon Agreement*).

PACIFIC REGION - GENERAL

Overview of Policies and Programs:
(http://www.pac.dfo-mpo.gc.ca/fm-gp/species-especes/salmon-saumon/pol/index-eng.html
Brief summaries of policies (e.g. Wild Salmon Policy, Allocation Policy, Improved Decision Making, and Selective Fishing), brief summary of programs and treaties.

Oceans Program: (http://www.dfo-mpo.gc.ca/oceans/oceans-eng.htm)
Integrated Coastal Management, Marine Protected Areas, Marine Environmental Quality, Oceans Outreach, Oceans Act.

Pacific Region – Fisheries Management Main Page:
(http://www.pac.dfo-mpo.gc.ca/fm-gp/index-eng.html)

Twitter: DFO Pacific @DFO_Pacific
Information about Fisheries and Oceans Canada in the Pacific Region

Compliance and Enforcement: (http://www.dfo-mpo.gc.ca/fm-gp/enf-loi/index-eng.htm)
Enforcement issues and strategies, "*Observe, Record, Report'*.

Treaty and Aboriginal Fisheries Programs:
(http://www.pac.dfo-mpo.gc.ca/abor-autoc/agreements-ententes-eng.html)
Aboriginal Fishing, AFS Agreements, Programs, Treaty Negotiations.

Recreational Fisheries: (http://www.pac.dfo-mpo.gc.ca/yukon/rec/index-eng.html)
General information for salmon fishing: licencing, species ID, restricted fishing areas, tagging and test fishing, DFO office locations; Fishing Notices.

Commercial Fisheries: (http://www.dfo-mpo.gc.ca/fm-gp/peches-fisheries/comm/index-eng.htm)
groundfish, pelagic and minor finfish, salmon and shellfish homepages; test fishing and selective fishing information; commercial fishery notices (openings and closures); Integrated Fishery Management Plans.

Integrated Fisheries Management Plans: (http://www.pac.dfo-mpo.gc.ca/fm-gp/ifmp-eng.html)
(Registration required to view page); Groundfish, Herring, Shellfish (invertebrates), Minor Finfish, Salmon, Archived Management Plans.

Fisheries Notices: (http://www-ops2.pac.dfo-mpo.gc.ca/fns-sap/index-eng.cfm?)
Want to receive fishery notices by e-mail? If you are a recreational sport licence vendor, processor, multiple boat owner or re-distribute fishery notices, register your name and/or company at the web-site address above. Openings and closures, updates, and other relevant information regarding your chosen fishery are sent directly to your registered email. It's quick, easy and free.

Salmon Test Fishery - Pacific Region
(http://www-ops2.pac.dfo-mpo.gc.ca/xnet/content/salmon/testfish/default.htm)
Definition, description, location and target stocks.

Licencing: (http://www.pac.dfo-mpo.gc.ca/fm-gp/licence-permis/index-eng.htm)
Contact information; Recreational Licencing Information, Commercial Licence Types, Commercial Licence Areas, Licence Listings, Vessel Information, Vessel Directory, Licence Statistics and Application Forms.

National On-line Licensing System (NOLS): (https://fishing-peche.dfo-mpo.gc.ca/)
E-mail: fishing-peche@dfo-mpo.gc.ca (include your name and the DFO Region in which you fish).

Salmon: (http://www.pac.dfo-mpo.gc.ca/fm-gp/species-especes/salmon-saumon/index-eng.html)
Salmon facts, salmon lifecycle, salmon fisheries, enhancement and conservation, research and assessment, salmon consultations (meeting schedules, minutes); policies, reports and agreements; glossary of salmon terms.

Regional Publications Online: (http://www.pac.dfo-mpo.gc.ca/publications/index-eng.html)
Agreements, treaties and acts, brochures and fact sheets, guideline documents, newsletters, reports and technical publications.

Ecosystems: (http://www.dfo-mpo.gc.ca/pnw-ppe/index-eng.html)
Working near water, publications (legislation, policy, guidelines, educational resources, brochures, newsletters and bulletins, papers and abstracts, reports).

***Species at Risk Act* (SARA) Legal Listings Consultations:** (www.sararegistry.gc.ca).
Information on the *Species at Risk Act* and how it is being implemented; Links to SARA public registry, COSEWIC, Canada Gazette; listing of current consultations.

Science: (http://www.dfo-mpo.gc.ca/science/index-eng.htm)
Science divisions, research facilities, PSARC, international research initiatives.

Community Mapping Network: (http://www.cmnbc.ca/)
Community Mapping Network (CMN) is to provide a service that collects and integrates natural resource information, maps and mapping information to promote sustainable resource management and to assist planning sustainable communities in the Yukon and British Columbia, Canada

Maps with Data & GIS Info: (http://www.pac.dfo-mpo.gc.ca/gis-sig/maps-cartes-eng.htm)
GIS Maps and Data

YUKON TRANSBOUNDARY RIVERS AREA

Fisheries and Oceans Canada, Yukon Transboundary Rives Area: (http://www.pac.dfo-mpo.gc.ca/yukon/index-eng.html)

YUKON RIVER PANEL

Yukon River Panel: (http://yukonriverpanel.com/salmon/)
Yukon River Salmon Agreement, Restoration and Enhancement Fund Reports, Panel reports.

YUKON SALMON SUB-COMMITTEE

Yukon Salmon Subcommittee Homepage: (http://www.yssc.ca/)
A public advisory body to the Minister of Fisheries & Oceans Canada and Yukon First Nations.

FORWARD

The purpose of this Integrated Fisheries Management Plan (IFMP) is to identify the main objectives and requirements for the Canadian Yukon River salmon fishery, as well as the management measures that will be used to achieve these objectives. This document also serves to communicate the basic information on the fishery and its management to Fisheries and Oceans Canada (DFO) staff, legislated co-management boards and committees (including the Yukon Salmon Sub-Committee), First Nations, harvesters, and other interested parties. This IFMP provides a common understanding of the basic "rules" for the sustainable management of the fisheries resource.

This IFMP is not a legally binding instrument that can form the basis of a legal challenge. The IFMP can be modified at any time and does not fetter the Minister's discretionary powers set out in the *Fisheries Act*. The Minister can, for reasons of conservation or for any other valid reasons, modify any provision of the IFMP in accordance with the powers granted pursuant to the *Fisheries Act*.

Where DFO is responsible for implementing obligations under Yukon First Nation Final Agreements, the IFMP will be implemented in a manner consistent with those obligations.

NEW FOR 2016/2017

Highlights / Key Changes for the Yukon River IFMP

Yukon Salmon Conservation Catch Card Available Online

- Anglers who plan to fish for salmon in the Yukon (Yukon River mainstem and all tributaries as well as the Alsek River watershed) may now purchase their Salmon Conservation Catch Card online (http://www.pac.dfo-mpo.gc.ca/yukon/rec/catchcard-carteprises-eng.html)

Chinook Salmon in the Yukon River Mainstem

- Outlook is for a weak run.
- Limited opportunities for First Nation subsistence fisheries are anticipated.
- Restrictions in recreational fishery are anticipated.
- No directed commercial or domestic fishery opportunities are anticipated.
- Updates to the in-season management matrix implemented.

Fall Chum Salmon in the Yukon River Mainstem

- Outlook is for a below-average run size for even-numbered years, but large enough to provide opportunities for Canadian fisheries.
- No restriction of First Nation subsistence fishery allocation is anticipated.
- No restriction of recreational fishery allocation is anticipated.
- Opportunity for directed commercial and domestic fishery is anticipated.

Chinook Salmon in the Porcupine River

- Outlook is for a weak run.
- Limited opportunities for the First Nation subsistence fishery is anticipated.

Fall Chum Salmon in the Porcupine / Fishing Branch River

- Outlook is for a weak run.
- A conservative approach is likely required to limit First Nation subsistence fishery harvest.

1. OVERVIEW

1.1 Introduction

Fisheries and Oceans Canada is responsible for the conservation and sustainable use of Canada's fisheries resources and is the principal management authority for Yukon River salmon. A number of governments and mandated bodies are also involved in the management of salmon harvest in Canadian portion of the Yukon River. The Yukon Salmon Sub-Committee (YSSC) of the Fish and Wildlife Management Board is established pursuant to Chapter 16 of each *Yukon First Nation Final Agreement*, as described in the framework *Umbrella Final Agreement* (UFA). The YSSC is established as the main instrument of salmon management in the Yukon and has the mandate to make recommendations, in the public interest, to the Minister of Fisheries and Oceans Canada and to Yukon First Nations on matters related to salmon. The YSSC is required to annually consult with Yukon First Nations and subsequently provide recommendations to the Minister of DFO on allocation of salmon by both user groups and areas. DFO supports the YSSC through the provision of technical expertise and participation in the First Nation consultation and public meeting processes. The engagements of the YSSC have been established as the principal form of First Nation, public and stakeholder consultations on all matters pertaining to Yukon River salmon in Canada.

When adult Canadian-origin Yukon River return to their natal streams to spawn, they migrate through Alaska before reaching Canada and their spawning grounds. There are fisheries on both sides of the international border. Given the transboundary (international) nature of the Yukon River, management of Canadian-origin salmon stocks are governed under Chapter 8 of the *Pacific Salmon Treaty* (PST) (*Yukon River Salmon Agreement*, YRSA). The YRSA is implemented through the bilateral U.S.-Canada Yukon River Panel which has the authority to provide recommendations on escapement goals, harvest sharing provisions and management measures to signatories to the Agreement. Consistent with *Yukon First Nation Final Agreements*, YSSC members comprise the majority of the Canadian members of the Yukon River Panel.

This IFMP, and management strategies described within, are based on recommendations from the Yukon River Panel, the YSSC and Yukon First Nations and covers a one-year span. It concerns the management strategies for Chinook and fall chum[1] salmon fisheries on the Yukon River. The IFMP contains comprehensive decision guidelines, which set out the rationale for management decisions and describes the range of departmental responses to changing in-season information. Decision guidelines may be reviewed and modified, if necessary, to reflect new considerations. This document also contains a brief overview of Yukon River salmon fisheries and is meant to inform fishers, processors and other interested parties about the expected run sizes, management considerations, and plans.

[1] Canadian-origin Yukon River chum salmon are referred to as "fall chum" salmon since their timing and genetic stock identification is distinct from "summer chum" salmon, which typically migrate earlier and spawn mostly within US portions of the drainage.

Management actions outlined in this plan are subject to change in response to in-season variables such as salmon migration timing, abundance, and environmental conditions. While fishing opportunities outlined in this plan are anticipated based on pre-season information, they are not guaranteed. The YSSC and DFO will continue to consult with First Nations, commercial, domestic[2] and recreational fishers throughout the season regarding fishing activities and allocations, particularly when in-season revisions are required to address specific conservation concerns.

The development and implementation of this IFMP supports the Departmental commitment to achieving long-term goals of salmon conservation, sustainable use of the resource and improved decision-making processes through consultation. Names, addresses and telephone numbers of the principal contact persons involved with Yukon River salmon management matters appear in the Contacts section of this document. Feedback on this IFMP is encouraged so that future plans can be made as useful as possible to stakeholders.

General information, such as details about biology, policy, historical information and many other departmental initiatives can be found online through a variety of web links listed in the Index of Web-Based Information section of this document.

1.2 History

Salmon have always played a pivotal role in the fabric of the Yukon. They are an integral part of the ecosystem providing a source of food and nutrients for a wide variety of flora and fauna. They have been a key food source for First Nations for millennia and more recently, played a very important part in the socio-economic life and in the developmental history of north-western Canada.

Because of their significance and the very high level of interest in ensuring the persistence of these populations, prudent and careful management supported by the broad spectrum of interests is required. Salmon are currently under a variety of threats including unstable conditions resulting from environmental changes, marine conditions, and in some cases, overexploitation.

1.3 Location of Fishery

This IFMP describes the Chinook and fall chum salmon fisheries in the Canadian portion of the Yukon River watershed (see Map Appendix 8).

1.4 Types of Fishery, Participants and Characteristics

There are four different types of fisheries for salmon in the Canadian portion of the Yukon River drainage including: First Nation (subsistence); commercial; recreational; and domestic[2] fisheries.

[2] The Yukon domestic fishery is a non-aboriginal subsistence fishery.

1.4.1 First Nation Fishery

The longest standing fishery in the Yukon is the First Nation fishery, which is widespread throughout the Yukon River drainage in Canada. Yukon First Nation fishers have traditionally relied heavily on the salmon resources of the Yukon. In accordance with DFO's approach to fishery prioritization in the Yukon, once conservation needs are met Yukon First Nation fisheries receive primary access to salmon. The First Nation subsistence fishery is managed through DFO's communal licensing process. In accordance with individual *First Nation Final Agreements*, First Nation individuals who wish to fish for subsistence purposes outside their traditional territory must first obtain consent from the First Nation whose territory they wish to fish in.

Currently 12 communal licences are issued annually to First Nations within the Yukon (including the Porcupine River) watershed. Subsistence fisheries primarily employ set gillnets, fish wheels (in larger tributary sites and in the mainstem Yukon River) drift gillnets (in the Teslin River), and gaffs in the smaller headwater streams. Depending on annual run timing, First Nation fishers usually commence fishing for Chinook salmon in early to mid-July and continue until subsistence needs are met. Fishing for fall chum salmon in the upper Yukon is usually completed by mid-October. However, on the Porcupine River, the fishery continues to operate through November with netting frequently occurring under the ice.

1.4.2 Commercial Fishery

The Canadian Yukon River commercial salmon fishery began in 1898. The commercial fishery currently involves up to 22 licensed fishers, with a minimum of eight additional licences guaranteed to Yukon First Nations (for a total of 30).

The mainstem Yukon River commercial salmon fishery is restricted to the following areas:

- In the Yukon River, downstream from Tatchun Creek to Dozen Islands (excluding a closed section around the mouth of the Klondike River);
- In the Stewart River downstream from the mouth of the McQuesten River until September 30, otherwise closed; and
- In the Pelly River downstream from the mouth of the MacMillan River.

Commercial fishing gear consists of fish wheels and gillnets. Fish wheels are only permitted in the mainstem Yukon portion of the commercial fishing areas.

During years of sufficient salmon abundance (when there are no conservation concerns), the commercial fishery typically opens in early July for Chinook salmon with specific schedules dependent upon run timing and the strength of the run. The commercial fishery for fall chum salmon occurs after Chinook (based on run timing), peaking in mid-September and concluding in mid- to late-October.

Commercial licences are administered via the web-based National Online Licensing System (NOLS). Through this system, commercial harvesters/licence holders/vessel owners may view, pay for, and print their commercial fishing licences, licence conditions and receipts. Licence renewal and payment of fees is mandatory on an annual basis prior to the expiry date of each fishery, in order to maintain the eligibility to be issued the licence in the future. Please note the licence eligibility will cease if it is not renewed annually.

For queries, NOLS access problems, or transactions that are not yet available in NOLS (e.g. vessel replacements and nominations), licensing services will continue to be available via:

Telephone:	1-877-535-7307	(request / identify 'Pacific Region')
Fax:	1-604-666-5855	
E-mail:	fishing-peche@dfo-mpo.gc.ca	(specify 'Pacific Region' in the subject line)

Please visit the Pacific Region Licencing website and subscribe to fishery notices for updates on NOLS and licencing services: http://www.pac.dfo-mpo.gc.ca/fm-gp/licence-permis/index-eng.html . Information on NOLS may be found on the DFO internet site at: http://www.dfo-mpo.gc.ca/fm-gp/sdc-cps/licence-permis-eng.htm.

1.4.3 Recreational Fishery

The first official recreational salmon fishing licences in the Yukon were issued in 1949 shortly after the construction of the Alaska Highway (1942-1947). The majority of recreational fishing effort for salmon occurs in the mainstem Yukon River near Tatchun River, as well as Klondike, Teslin, and Pelly rivers. The timing of recreational Chinook salmon fishing effort on the Yukon River is typically from mid- to late-July through August. It is unlawful to use any hook other than a single-pointed barbless hook with a distance of 2 cm (3/4 in.) or less between the point and the shank while angling for salmon in the Yukon River and major tributaries from July 1st to October 15th. Fall chum salmon are not commonly targeted in the recreational fishery.

In addition to holding a valid Yukon Angling Licence, anglers targeting Yukon River salmon are also required to obtain a Yukon Salmon Conservation Catch Card (Catch Card) prior to fishing for salmon. If a salmon is caught and landed, the angler must immediately record the date, location, species, sex, presence of tags, presence of adipose fins and type of gear used. This information must be recorded even if the salmon is intended to be (or is) released. All recreational anglers who are issued a Catch Card must return the completed card (even if salmon fishing did not occur, or no salmon were caught) to DFO by no later than November 30. Failure to submit the Catch Card could result in a fine or forfeiture of future recreational salmon fishing privileges. Catch Cards are available for purchase online, through the National Recreational Licencing System (NRLS: https://www-ops2.pac.dfo-mpo.gc.ca/nrls-sndpp/index-eng.cfm) as well as a number of over-the-counter vendors throughout the Yukon.

1.4.4 Domestic Fishery

The domestic fishery was first initiated in 1899 to allow British subjects and Yukon residents to fish for personal use with up to 300 yards of gillnet. This fishery was eliminated in 1961 but then re-instated in 1974 to allow Yukon residents living in remote areas to harvest salmon for food. The domestic fishery currently involves 7 licensed fishers who fish primarily for Chinook salmon. Domestic fisheries are restricted to the same geographic areas as commercial fisheries while fishing gear is limited to one gillnet of up to 90 metres in length. When there are no conservation concerns, the domestic fishery generally follows the same schedule of openings and closures as the commercial fishery.

2. GOVERNANCE

Departmental policy development related to the management of fisheries is guided by a range of considerations including legislated mandates, judicial guidance, international and domestic commitments, and a precautionary, ecosystem-based approach to the management of resources. This section provides a brief overview of key policies and the legal context for Pacific salmon management. Policies are developed with considerable consultation from all those with an interest in salmon management. While the policies themselves are not subject to annual changes, implementation details are continually refined where there is general support.

Additional information is accessible on-line and can be found through the Index of Web-based Information in this report.

2.1 Pacific Salmon Treaty

In March 1985, Canada and the United States agreed to co-operate in the management, conservation, harvest sharing, research, and enhancement of Pacific salmon stocks of mutual concern by ratifying the Pacific Salmon Treaty (PST). The Pacific Salmon Commission (PSC), established under the PST, provides regulatory and policy advice as well as recommendations to Canada and the United States with respect to interception salmon fisheries.

Although fishing arrangements pertaining to Yukon River salmon stocks were not initially included in the PST when it took effect in 1985, it did commit the Parties (i.e., Canada and the United States) to negotiate arrangements for Canadian-origin Yukon River salmon. Between 1985 and 2001 Canadian and U.S. delegations, which included First Nation representatives, fishers, processors and government representatives met frequently to negotiate. On March 29, 2001, negotiators initialed a draft Yukon River Salmon Agreement (YRSA) intended to address key matters pertaining to Canadian-origin salmon in the Yukon River issues. Highlights of the agreement include: the creation of the Yukon River Panel (YRP) and Yukon River Joint Technical Committee (JTC); conservation and harvest sharing targets for Canadian-origin salmon stocks; guidance for coordinated management (including spawning requirements), rebuilding plans, habitat protection and restoration and enhancement guidelines; and, the Yukon River Restoration and Enhancement Fund. With an exchange of diplomatic notes in December 2002, the Parties ratified the YRSA as Chapter 8 of the PST.

2.2 Umbrella Final Agreement and Yukon First Nation Final Agreements

The Umbrella Final Agreement (UFA) was approved in 1993 by the Government of Canada, Government of Yukon and Yukon First Nations as represented by the Council of Yukon First Nations (CYFN). The UFA has served as a framework for the establishment of 11 individual Yukon First Nation Final Agreements ratified in the Yukon to date. Yukon First Nation Final Agreements represent an exchange of undefined aboriginal rights for defined treaty rights. Individual Yukon First Nation Final Agreements set out specific rights for the particular First Nation and its Citizens which are protected under the Canadian Constitution. Chapter 16 of the Final Agreements establishes the framework for many aspects of salmon management and allocation processes in the Yukon, including the creation of the Yukon Salmon Sub-Committee and guaranteeing that the majority of Canadian representation on the Yukon River Panel is comprised of Salmon Sub-Committee members. First Nation access to Yukon River salmon for subsistence harvest purposes is afforded the highest priority after conservation requirements are met.

2.3 First Nations and Canada's Fisheries

The Government of Canada's legal and policy frameworks identify a special obligation to provide First Nations the opportunity to harvest fish for food, social, and ceremonial (FSC) purposes. The *Aboriginal Fisheries Strategy* (AFS) was implemented in 1992 to address several objectives related to First Nations and their access to the resource, including:

- Improving relations with First Nations
- Providing a framework for the management of the First Nations fishery in a manner that was consistent with the 1990 Supreme Court of Canada Sparrow decision
- Greater involvement of First Nations in the management of fisheries
- Increased participation in commercial fisheries (Allocation Transfer Program)

Where First Nation treaties have not been finalized, the AFS continues to be the principal mechanism that supports the development of relationships with First Nations including consultation, planning and implementation of fisheries, and the development of capacity to undertake fisheries management, stock assessment, as well as stock and enhancement programs.

The Aboriginal Aquatic Resources and Oceans Management (AAROM) program has been implemented in some areas to fund aggregations of First Nation groups to build the capacity required to coordinate fishery planning and program initiatives. AAROM is focused on developing affiliations between First Nations to work together at a broad watershed or ecosystem level – a level at which there is a certain number of common interests and where decisions and solutions can be based on integrated knowledge of several Aboriginal communities. In the conduct of their activities, AAROM bodies are working to be accountable to the communities they serve, while working to advance collaborative relationships between member communities, DFO and other interests in aquatic resource and oceans management.

2.4 Policy Framework for the Management of Pacific Salmon Fisheries

Salmon management programs continue to be guided by the following policies: *Canada's Policy for Conservation of Wild Pacific Salmon (Wild Salmon Policy)*, *An Allocation Policy for Pacific Salmon*, Pacific Fisheries Reform, *A Policy for Selective Fishing*, *A Framework for Improved Decision Making in the Pacific Salmon Fishery*, and the *Pacific Region Fishery Monitoring and*

Reporting Framework.

The *Wild Salmon Policy* (WSP), which was approved in 2005, sets out a process for the protection, preservation and rebuilding of wild salmon and their marine and freshwater ecosystems for the benefit of all Canadians. The policy provides for the identification of irreplaceable groupings of stocks, called Conservation Units (CUs), and the identification of upper and lower benchmarks that are a measure of the status of each CU. Other features of the WSP include the monitoring of habitat status and a process for public engagement in the establishment of long term strategic plans for Conservation Units. In implementing the WSP, the Department has focused on the identification of Conservation Units following a standardized methodology, which included the examination of the distribution and timing of salmon, genetics, eco-regions and climate.

For Canadian-origin Yukon River Chinook salmon, 12 Conservation Units have been identified:

1. North Yukon River (including the Yukon River and tributaries downstream of the Stewart-Yukon confluence)
2. Mid Yukon River (extending from the White-Yukon confluence upstream (and including) the Little Salmon drainage
3. Upper Yukon and tributaries upstream of the Yukon-Teslin confluence
4. Stewart River and tributaries
5. White River and tributaries
6. Pelly River and tributaries
7. Nordenskiold River and tributaries
8. Big Salmon River and tributaries
9. Teslin River including tributaries and headwaters
10. Old Crow River (tributary to Porcupine River)
11. Salmon Fork River (tributary to Porcupine River)
12. Porcupine River and all other tributaries (excluding Old Crow and Salmon Fork Rivers)

For chum salmon, 7 Conservation Units have been identified:

1. North Yukon River downstream from the Yukon-White confluence
2. Middle Yukon River
3. South Yukon River upstream of the Yukon-Teslin confluence
4. White River and tributaries
5. Teslin River and tributaries
6. Porcupine River and tributaries including the Fishing Branch
7. Old Crow River (tributary to the Porcupine River)

In the future, the Department anticipates a review and renewal of the Wild Salmon Policy Implementation Plan in order to allow alignment with changes to legislation and programs since its release in 2005.

An Allocation Policy for Pacific Salmon, announced in 1999, contains principles to guide the management and allocation of the Pacific salmon resource between First Nations, commercial and recreational harvesters, and forms the basis for general decision guidelines outlined in this plan.

2.5 Fishery Monitoring and Catch Reporting

A complete, accurate, and verifiable fishery monitoring and catch reporting program is required to successfully balance conservation with the objectives of optimal harvest levels. Across all fisheries, strategies exist or are being developed to improve catch monitoring programs by identifying standards that must be achieved as well as by clarifying roles and responsibilities of the Department and harvesters. Catch monitoring programs have been developed for Yukon fisheries including First Nation (subsistence), domestic, recreational and commercial. Monitoring programs will ensure that the fishery information required to make critical management decisions is available to all those who need it, when it is required. Furthermore, catch data are required to effectively manage and report on domestic and international harvest sharing arrangements. The Department finalized the "Strategic Framework for Fisheries Monitoring and Catch Reporting in the Pacific Fisheries" (the Framework) in the spring of 2012. The Framework outlines how consistent risk assessment criteria can be applied to each fishery to determine the level of monitoring required, while allowing for final monitoring and reporting programs to reflect the fishery's unique characteristics. More information is available at:
www.pac.dfo-mpo.gc.ca/fm-gp/docs/framework_monitoring-cadre_surveillance/page-1-eng.html

2.6 Species at Risk Act

The *Species at Risk Act* (SARA) came into force in 2003. The purposes of the Federal Act are "*to prevent wildlife species from being extirpated or becoming extinct, and to provide for the recovery of a wildlife species that is extirpated, endangered or threatened as a result of human activity and to manage species of special concern to prevent them from becoming endangered or threatened".* By definition, an extirpated wildlife species is a species that no longer exists in the wild in Canada but exists elsewhere in the wild, whereas, an extinct wildlife species is a wildlife species that no longer exists.

In addition to the existing prohibitions under the *Fisheries Act*, it is illegal to kill, harm, harass, capture, take, possess, collect, buy, sell or trade any species listed under SARA or any part or derivative of an individual of a SARA listed species. It is also illegal to damage or destroy a listed species residence. These prohibitions apply unless a person is authorized, by a permit, licence or other similar document issued in accordance with SARA, to engage in an activity affecting the listed species or the residences of its individuals.

In the Yukon, no salmon species are currently listed under SARA. More information on SARA can be found at www.sararegistry.gc.ca/ .

2.7 Salmonid Enhancement Program

Fisheries and Oceans Canada's Salmonid Enhancement Program (SEP) in the Pacific Region is comprised of nearly 300 projects including hatcheries, fishways, spawning and rearing channels, and small classroom incubators. Projects range in size from spawning channels producing nearly 100 million juvenile salmon annually to school classroom incubators releasing fewer than one hundred juveniles (per aquarium). SEP enhances Chinook and chum salmon at the population level supporting sustainable fisheries through fish production that provides harvest opportunities. Fish production from the program also supports stock assessment and conservation, both of which enable harvest management as well as community involvement and public education. Although the vast majority of SEP projects are located in British Columbia, a few existing (and emerging) projects have been fostered in the Yukon.

2.8 Scientific Support

Research on Yukon River salmon stocks is being conducted through coordinated efforts of the Department, U.S. agencies (Alaska Department of Fish and Game, United States Fish and Wildlife Service, and the National Marine Fisheries Service, Yukon First Nations and a number of private firms and academic institutions. Many of the public programs and some government projects are funded through the Yukon River Restoration and Enhancement Fund.

The annual report of the Yukon River Panel's Joint Technical Committee provides a comprehensive summary and data report on research and management-related activities undertaken on the Yukon River. The most recent JTC report can be accessed through the YRP website at: http://yukonriverpanel.com/salmon/publications/joint-technical-committee-reports/

Some of the research activities of the Department's Science Sector are summarized in annual reports and/or scientific papers that are peer reviewed through Centre for Scientific Advice – Pacific (CSAP). The advice is then forwarded to for review and adoption as required. Additional information and reports are available at: (http://www.pac.dfo-mpo.gc.ca/fm-gp/species-especes/salmon-saumon/research-recherche/assessment-eng.html)

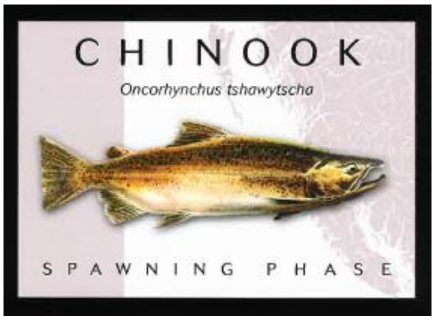
2.9 Integrated Fisheries Management Plan Approval Process

Following the development of this IFMP by the Department's Yukon Transboundary Rivers Area operational office, the plan is reviewed by senior Departmental officials with responsibility for salmon management in Pacific Region. The IFMP is considered final once approved by the Pacific Regional Director General of Fisheries and Oceans Canada.

3. STOCK ASSESSMENT, SCIENCE and TRADITIONAL ECOLOGICAL KNOWLEDGE

3.1 Biological Synopsis

3.1.1 Chinook Salmon (Oncorhynchus tshawytscha)

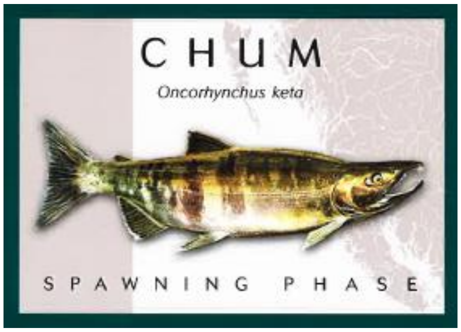


Chinook salmon spawn in streams and rivers along the west coast of North America. The Yukon River is one of the most northerly of the major Chinook spawning rivers, and hosts some of the longest upstream migrating salmon stocks in the world. Some headwater stocks migrate in excess of 2,960 kilometres in freshwater to reach their spawning grounds in the Yukon and northern British Columbia. The majority of Chinook salmon spawning in the upper Yukon River occurs in August.

Over the winter, the eggs incubate in the gravel and Chinook fry emerge in the spring and early summer. Some fry leave the rivers of their birth, or "natal" streams, soon after emergence. They may be carried downstream into larger rivers by the spring freshet. Through the summer, many fry will migrate into "non-natal" streams to feed and may migrate significant distances upstream (in a number of documented cases upwards of 75 kilometers) and hundreds of kilometers downstream. In large rivers, juvenile Chinook are often found along the river margins and in the mixing zones where streams and rivers join larger ones. In lakes, Chinook fry have been found in nearshore habitats and near the mouths of tributaries. Juveniles may be abundant in tributaries not used for spawning.

Yukon River Chinook salmon fry must grow rapidly and build up reserves of fat for their first winter in freshwater. This is called the freshwater "rearing" phase. Successful over-wintering of juveniles has been documented only in streams and smaller rivers, although it is expected to also occur in larger rivers. After their winter in freshwater as free swimming juveniles, fry begin their downstream migration to the ocean. The first winter at sea is thought to be a very important time for Chinook salmon and survival during this period can greatly influence the strength of this brood year. Chinook salmon then spend the next 2 to 5 years in the Bering Sea before returning to their natal spawning grounds. Most Chinook salmon return as 5 year olds or 6 year olds, but some return as 4 year olds or 7 year olds (typically less than 10%). There used to be some Chinook salmon in the Yukon River that returned as 8 year olds but none have been recorded for several decades.

3.1.2 Fall Chum Salmon (*Oncorhynchus keta*)



Fall chum salmon spawn in rivers and streams along most of the west coast of North America, and along the Bering and Arctic coasts eastwards to, and including, the Mackenzie River drainage. The upper Yukon River stocks of this species may have the longest upstream fall Chum spawning migration in the world with some stocks migrating over 2,700 kilometres[3] in freshwater. In more southerly rivers, adult freshwater migrations tend to be much shorter with spawning occurring closer to the estuaries.

There are two runs of chum salmon that enter the mouth of the Yukon River. The first to arrive are the "summer chum", which enter the river mouth in early June and reach peak abundance around the third week of June. Summer chum generally spawn in the lower 800 kilometres of the Yukon drainage and, until recently, were not believed to migrate into the Canadian section of the drainage. The observance of early-timed fall chum salmon found in the Chandindu River, Mickey Creek (tributary of the Fortymile River), and the Klondike River could possibly be a component of the larger Yukon River "summer chum" stock.

Adult fall chum salmon are characterized at the river mouth by later run timing, larger body size and a more silvery appearance than summer chum salmon. Spawning occurs primarily in the upper portions of the drainage. Upper Yukon fall chum salmon return as spawning adults to the river mouth generally from mid-July through early September after spending up to five years in the ocean. Peak migration timing of fall chum salmon entering the Canadian portion of the drainage usually occurs mid-September. Predominant age classes of mature upper Yukon fall chum salmon include age-four (62%) and age-five (35%). Spawning has been documented in ground water discharge areas that have water with a fairly constant flow rate and temperature (between three and seven degrees Celsius), along cutbanks and in riffle areas of the mainstem Yukon, and in side channels and sloughs. Peak spawning usually occurs during October through early November.

Juvenile upper Yukon fall chum salmon emerge from the gravel during the spring (April/May). After emergence, they appear to spend little time in the natal area. By mid-June, most have moved away from the spawning areas however little is known about the downstream migration of juveniles although it likely occurs close to the time of spring break-up.

[3] Spawning fall chum salmon have been observed spawning on the Teslin River near Boswell Creek and are known to migrate into Teslin Lake.

3.2 Ecosystem Interactions

As a consequence of their anadromous life history, salmon are sensitive to changes in both the marine and freshwater ecosystems. Salmon are an ecologically important species supporting vast food webs in oceanic, estuarine, freshwater, and terrestrial ecosystems.

DFO is moving away from management of salmon by single species and moving towards an integrated ecosystem approach to science. Strategy 3 of *Canada's Policy for the Management of Wild Pacific Salmon*, also known as the Wild Salmon Policy or WSP, outlines the Department's intent to progressively incorporate ecosystem values in salmon management and identifies these actions:

- Identify indicators (biological, physical and chemical characteristics) to use in monitoring the status of freshwater ecosystems, and
- Monitor annual variation in climate and ocean conditions, integrate the monitoring with assessments of marine survival of Pacific salmon, and incorporate this knowledge into the annual forecasts of salmon abundance and management processes.

One of the greatest challenges in implementation of the WSP is balancing the goals of maintaining and restoring healthy and diverse salmon populations and their habitats, with social and economic objectives that reflect people's values and preferences. Standardized monitoring and assessment of wild salmon populations, habitat and eventually ecosystem status will facilitate the development of comprehensive integrated strategic plans (WSP Strategy 4) that will address the goals of the WSP while addressing the needs of people. Outcomes of these plans will include biological objectives for salmon production from Conservation Units and, where appropriate, anticipated timeframes for rebuilding, as well as management plans for fisheries and watersheds, which reflect open, transparent, and inclusive decision processes involving First Nations, communities, environmental organizations, fishers and governments.

3.3 Aboriginal Traditional Knowledge/Traditional Ecological Knowledge

Both Aboriginal Traditional Knowledge (ATK) and Traditional Ecological Knowledge (TEK) reflect the cumulative knowledge gathered over generations and encompass regional, local and spiritual connections to ecosystems and all forms of plant and animal life. ATK is knowledge held by Aboriginal peoples and communities, while TEK is local knowledge held by Non-Aboriginal communities, including industry, academia, and public sectors. While qualitatively different, both are regionally and locally specific and can often contribute to improving management.

The growing awareness of the value of ATK and TEK is reflected in the increasing requirements for both to be included in environmental assessments, co-management arrangements, species at risk recovery plans, and all coastal management decision-making processes. ATK and TEK may inform and fill knowledge gaps related to the health of salmon stocks and to aid decision making related to development and resource use. Government and the scientific community acknowledge the need to access and consider ATK and TEK in meaningful and respectful ways. A challenge for resource managers is how to engage knowledge holders and ensure that the information can be accessed and considered in a mutually acceptable manner, by both knowledge holders, and the broader community of First Nations, stakeholders, managers, and policy makers involved in the fisheries.

The WSP acknowledges the importance of integrating ATK and TEK into the strategic planning process. The Department is exploring best practices to develop an approach for incorporating ATK and TEK into WSP integrated planning. The Department will also consider identifying potential partnerships with First Nation organizations to develop an approach for integrating ATK into WSP, particularly in planning initiatives.

The federal SARA makes a special reference to the inclusion of Traditional Knowledge in the recovery of species at risk. The Department has developed an operational guidance document for SARA practitioners (*Guidance on Considering Traditional Knowledge in Species at Risk Implementation, 2011*).

3.4 Stock Assessment

Since 2009, a sonar program located at Eagle, Alaska (immediately downstream of the Canadian border) has been the main means of assessing both Chinook and fall chum salmon returning to the Yukon River (mainstem) in Canada and provides the border passage estimates necessary to manage mainstem Canadian fisheries. This sonar program replaced a long-standing mark-recapture assessment project. In the Porcupine River drainage, both a sonar program on the Porcupine River provides information on return of Chinook and chum salmon to the Canadian Porcupine watershed and a weir on the Fishing Branch River provides information on escapement of fall chum salmon to the drainage's main spawning grounds.

In addition to these border passage assessments there are several other assessment programs that provide information on spawning escapement in select tributaries. Aerial surveys of select index areas in the upper Yukon River are conducted in some years. The Whitehorse Rapids Fishway provides information on the escapement of wild and hatchery-origin returns into the upper Yukon drainage above the Whitehorse hydroelectric dam. Additional escapement enumeration projects are conducted by other parties including First Nations and independent contractors supported by the Yukon River Restoration and Enhancement Fund. Most catch and escapement monitoring programs also include a sampling component to determine the age, size and sex composition of the fish being monitored.

Considerable effort is being spent on collecting tissue samples from major spawning populations throughout the Yukon drainage to complete the genetic baselines for genetic stock identification (GSI) and to increase the capability to monitor specific stocks and/or groups of stocks that lack escapement data. GSI baseline sampling is being incorporated into many existing stock assessment projects (both agency and non-agency funded projects) throughout the drainage. The fall chum salmon genetic baseline is considered to be well developed, whereas the Chinook salmon baseline requires considerable effort due to the many individual stocks (approximately 100) and remote location of many of these stocks.

3.5 Precautionary Approach

Generally, scientific advice to Fisheries Managers considers data quality and incorporates uncertainty (i.e., stock status forecasts presented as a statistical distribution rather than point estimate). WSP benchmarks of biological status will inform the continuation of the precautionary approach to management of salmon resources. Decisions on recovery and fisheries objectives will consider the Strategic Planning Process described under WSP Strategy 4.

3.6 Research

An overview of the science & research in the Pacific Region is available on DFO's Pacific region website (see index of web-based information). Information on many Yukon River research projects can be found on the Yukon River Panel's website at:
http://yukonriverpanel.com/salmon/publications/reports/

Some current research projects on Yukon River salmon:

- Juvenile Chinook salmon investigations on the Big Salmon River. This study aims to understand the growth, movement, and production of Chinook salmon in the Big Salmon River.
- Porcupine River Habitat Investigations. This study aims to understand the current state of chum salmon habitat in the Fishing Branch River and to elucidate whether changed or degraded habitat may be the cause of the recent declines seen in this salmon stock.
- Pelly River sonar. This study aims to count the number of Chinook salmon returning to the Pelly River system in 2016.

4. SOCIAL CULTURAL AND ECONOMIC IMPORTANCE

4.1 Aboriginal Participation

First Nation culture recognizes the importance of stewardship and responsibility to care for salmon, a responsibility that has been handed down over time. Part of this stewardship responsibility is to ensure that salmon are available for future generations. Through their fishing activities, First Nation communities are able to maintain a linkage to the salmon and gain knowledge of the salmon stock's abundance and health. This continued awareness allows First Nation people to contribute to supporting the development of effective management strategies through the provision of information on local and regional observations. Consultation and engagement with First Nations includes participation on a number of levels and in a variety of ways. These exchanges and involvement may include bilateral consultations, advisory processes, management boards, technical groups and other roundtable forums.

Generally, DFO manages aboriginal fisheries to provide access for food, social, ceremonial (FSC) and for commercial purposes. With respect to fishing for FSC purposes, DFO manages this fishery to ensure that after conservation needs are met, the FSC fishery has priority over other fisheries.

4.2 Recreational Sector

Economic benefits from the recreational fishers include, but are not limited to the purchase of: angling licences, Salmon Conservation Catch Cards, angling equipment, accommodation, travel, and air charter services. In addition to economic benefits, recreational fishing also has added social and cultural benefits as it is considered a tradition and lifestyle for many people. Fishing provides people with the opportunity to interact with the natural environment and increases their awareness of salmon resources. The increased awareness is commonly associated with an enhanced sense of stewardship as well as overall social value.

4.3 Commercial Sector

Commercial fishers benefit from the salmon fishery economical, socially and culturally. The economic elements are often assessed through financial gains associated with commercial fishing activities, although the social and cultural benefits are not as easily quantified. Fishers may also derive benefits from the social aspects of the fishery, such as interactions with other fishers and fishery managers.

5. MANAGEMENT ISSUES

5.1 Canada – U.S. International Agreement: *Yukon River Salmon Agreement*

In March 2001, Canada and the U.S. concluded the Yukon River salmon negotiations, which had been ongoing since 1971. The comprehensive Yukon River Salmon Agreement (YRSA) was ratified by the Parties and incorporated into the PST in 2002. The YRSA contains specific obligations for the Parties to manage fisheries to achieve conservation and harvest sharing objectives for Canadian-origin Yukon River Salmon stocks.

5.2 Uncertainty Regarding Runs

There have been significant swings in the production of Yukon River salmon in recent years with generally poor runs in the 2007 to 2015 period for upper Yukon Chinook salmon, the 1998-2001 period for upper Yukon fall chum salmon and since 2008 for Fishing Branch River / Porcupine fall chum salmon. It is believed that changes in marine survival play a significant role in these abundance fluctuations and it is reasonable to expect these situations may continue in the foreseeable future. Changes in marine conditions have proven challenging to monitor and forecast while the resulting effects on salmon survival and production has been difficult to predict.

5.3 Uncertain Market Conditions

Low fish volumes and the lack of a major local buyer/processor have hindered the commercial fishery for over the past decade. Although the opportunity to harvest fall (mainstem) chum salmon is anticipated in 2016, the availability of viable markets due to remoteness and timing of the fishery is anticipated to continue to be a significant limiting factor.

6. OBJECTIVES

6.1 Conservation Objectives

Restore and maintain healthy and diverse salmon populations and their habitat for the benefit and enjoyment of the people of Canada in perpetuity.

Fisheries will be managed in accordance with the WSP. The policy goal listed above will be advanced by safeguarding the genetic diversity of wild salmon populations, maintaining habitat and ecosystem integrity, and managing fisheries for sustainable benefits.

The fisheries management approach defined within the Yukon River Salmon Agreement of the PST is abundance-based. This approach defines resource conservation as the paramount objective, with harvest fluctuating according to actual abundance rather than to pre-determined (guaranteed) levels. Abundance-based management (ABM) approaches have been developed for upper Yukon Chinook and fall chum salmon as well as Porcupine (Fishing Branch) River fall chum salmon.

DFO establishes escapement goals for the Chinook and fall chum salmon returns prior to each fishing season after considering recommendations from the YRP and YSSC. The Yukon River In-Season Fishery Management Decision Matrices were developed following extensive consultations with Yukon First Nations, fishers, the YSSC. These matrices provide pre-defined specific reference (trigger) points and associated management actions. The trigger points are separated into three management zones: the RED ZONE where no harvesting opportunities are available; the YELLOW ZONE where reductions to First Nation subsistence fisheries are implemented and no harvest opportunities through other fisheries exist; and, the GREEN ZONE where fishing opportunities for all fisheries are considered. Conservation concerns are foremost in years with low run sizes.

For 2016, the YRP adopted the following spawning escapement goal ranges:

Chinook salmon – Yukon River Mainstem in Canada	42,500 – 55,000
Fall chum salmon – Yukon River Mainstem in Canada	70,000 – 104,000
Fall chum salmon – Fishing Branch River (Porcupine Drainage) in Canada	22,000 – 49,000

6.2 First Nation Fisheries Objectives

The objective is to manage fisheries in recognition of *Section 35* of the Canadian Constitution, constitutional priority of the First Nations fishery.

Subject to conservation needs, first priority is afforded to First Nations for opportunities to harvest Yukon River salmon for FSC / subsistence purposes and any treaty obligations. Specific treaty obligations and considerations are described within individual First Nation Final Agreements.

6.3 International Objectives

The objective is to manage Canadian fisheries on the Yukon River to ensure that obligations within the YRSA are achieved.

Besides meeting the escapement, management, allocation and conservation targets described in this IFMP, Canada has an overarching international obligation to manage its harvest within agreed harvest sharing arrangements as defined within the YRSA.

6.4 Domestic Allocation Objectives

The objective is to manage fisheries in a manner that is consistent with *An Allocation Policy for Pacific Salmon*.

An Allocation Policy for Pacific Salmon can be found on-line at: http://www.pac.dfo-mpo.gc.ca/fm-gp/species-especes/salmon-saumon/pol/index-eng.html. The *Allocation Policy for Pacific Salmon* identifies the priority for allocation of salmon harvest amongst fisheries. The allocation priorities are described below:

Priority 1: Attain escapement goals and maintain fish habitat that will result in optimum production of the stocks;

Priority 2: Provide for the basic needs of the First Nation fisheries for FSC purposes, and for the Basic Needs Allocations, where specified, in First Nation Final Agreements; and

Priority 3: Provide salmon harvesting opportunities for recreational, commercial and domestic fishers. Relative to the commercial and domestic fisheries, recreational fisheries are provided priority access to Chinook and coho and predictable and stable fishing opportunities for chum salmon. These fisheries are provided opportunities to harvest fish providing there are no restrictions required in the First Nation fishery.

International allocations are specified in the YRSA, whereas domestic allocations may be recommended by the YSSC, in consultation with stakeholders. These recommendations are frequently influenced by the historical performance of respective fisheries.

Achieving these objectives in the Yukon drainage is difficult due to many factors such as: the biological complexity of the stocks, the wide distribution of spawning streams, wide fluctuations in run sizes, increasing efficiency and demands of user groups, and the requirement for a precautionary approach to fisheries management that protects and conserves wild stocks.

6.5 Communication Objectives

The objective is to provide timely information to fishers, communities and the public regarding the status of salmon runs and management decisions.

Fisheries and Oceans Canada will compile and communicate weekly run status updates once Chinook salmon arrive at the international border. Included in the updates will be the latest stock assessment information based on the Eagle sonar assessment program, catch information and fishery management information (i.e. openings/closures) in both Canada and the U.S..

6.6 Enforcement Objectives

The objective is to ensure compliance with Acts and Regulations associated with the management of Pacific salmon.

The *Yukon Territory Fishery Regulations*, the *Fishery (General) Regulations* and the *Aboriginal Communal Fishing Licences Regulations*, established pursuant to the *Fisheries Act*, are the main pieces of legislation for salmon fisheries in the Yukon. The Conservation and Protection (C&P) program of Fisheries and Oceans Canada is responsible for monitoring and enforcing compliance with the *Fisheries Act* and the associated regulations in relation to anadromous fish in both lakes and river systems, and to ensure compliance with habitat provisions in all water frequented by fish. C&P will continue to work with cooperatively with First Nations and other Federal and Territorial agencies and departments to deliver services.

Fishery officers work closely with other management and enforcement agencies such as the Canadian Food Inspection Agency, RCMP, Conservation Officers, Parks Canada, Canada Border Services Agency and First Nation management bodies. These partners assist fishery officers in carrying out their mandate. Where possible, the sharing of human resources and equipment reduces the occurrence of overlapping patrols and shows a concerted effort to manage cost-effectively within a budget. Due to the remoteness and extensive size of the patrol area, patrol efforts are undertaken in a strategic manner based on pre-identified priorities or, where appropriate, as a result of complaints or identified concerns rather than on random/routine patrols.

7. ACCESS AND ALLOCATION

Allocation decisions are made in accordance with the recommendations from the YSSC, First Nation Final Agreements and the *Allocation Policy for Pacific Salmon*. The allocation policy is based upon a hierarchy of priorities. At low run sizes, subject to conservation concerns, the only fisheries that are provided an allocation are First Nations' fisheries for FSC purposes. At higher run sizes, fishing opportunities for recreational, domestic, and commercial fisheries will be considered as long as the projected run abundance is sufficient to meet escapement and First Nation requirements. The BNA of chum, Chinook, and coho salmon for the Vuntut Gwitchin First Nation (VGFN) in the Porcupine River is specified in the VGFN Final Agreement (16.10.7.4). The Basic Needs Allocations (BNA) for First Nations that harvest mainstem Yukon River salmon have yet to be finalized. Nonetheless, a primary objective of this management plan is to address the requirements of the First Nation fisheries in Yukon.

7.1 Long Term Objectives for the Fisheries

There are four key long-term objectives:

a) Meet the obligations contained in the Canada/U.S. *Yukon River Salmon Agreement*

b) Conserve and restore spawning stocks and habitats

c) Determine the basic needs of the First Nation fishery

d) Develop and/or maintain sustainable and viable Canadian fisheries

7.1.1 Meeting Obligations of the Yukon River Salmon Agreement

The YRP met twice during the 2015/16 meeting cycle, once in December 2015 (in Whitehorse) and once in April 2015 (in Anchorage). Principal items addressed by the YRP included a post-season review of the 2015 salmon season, reviewing the 2016 run outlooks, reviewing and approving projects to be funded under the Restoration and Enhancement Fund in 2016 and recommending escapement goals for Canadian-origin Yukon River salmon stocks.

The Yukon Joint Technical Committee met twice during the past year and prepared the *Yukon River Salmon 2015 Season Summary and 2016 Season Outlook* annual report. The report is available through Fisheries and Oceans Canada, the Alaska Department of Fish and Game or the YRP website (http://yukonriverpanel.com/salmon/wp-content/uploads/2009/03/rir3a201601.pdf).

Since 2002, Canadian Yukon River salmon management plans have been developed to ensure consistency with harvest sharing provisions of the YRSA. Extensive communications occur during each season between Canadian and U.S. fishery managers to exchange updated in-season data and fishery information. Through funding from the Restoration and Enhancement Find, the Yukon River Drainage Fisheries Association (YRDFA) will again be hosting drainage-wide (U.S. / Canada) conference calls to gather and disseminate information about fishery and run status throughout the season. The calls are intended for individuals from communities along the Yukon River and are held once a week from June through to August. DFO also hosts a weekly conference calls with Canadian First Nation representatives. The purpose of the weekly calls is to disseminate information on the status of Canadian-origin salmon runs, assessment programs, and the in-season management strategy.

7.1.2 Conserving and Restoring Salmon Spawning Stocks and Habitat

Management measures taken to address conservation concerns for Yukon salmon stocks include closures and/or delayed openings of the commercial, domestic, and recreational fisheries. If conservation concerns exist, these fisheries are generally not opened until in-season assessments indicate that escapement targets and First Nation requirements will be achieved. Reduction or removal of the total allowable catch allocations to First Nation subsistence fisheries will occur through receipt of recommendations from the YSSC or in emergency situations as described within Yukon First Nation Final Agreements.

7.1.3 Developing and/or Maintaining Sustainable and Viable Canadian Fisheries

Progress on this issue has been accomplished through the implementation of an abundance-based management approach and through some of the harvest sharing, run rebuilding and restoration and enhancement provisions negotiated in the YRSA. As has become apparent during recent years of low productivity, the sustainability and/or viability of some fisheries at the levels maintained prior to 1998 is not likely achievable in the foreseeable future. During times when commercial harvest opportunities are low, greater effort will be required to maximize the benefit from the fish that can be harvested. This may require new approaches to the handling and processing of salmon and the development of innovative value-added approaches.

8. RUN OUTLOOKS, DECISION GUIDELINES, AND MANAGEMENT MEASURES

A comprehensive overview of the pre-season run forecasts and outlooks can be found in section 10 of the Joint Technical Committee's *Yukon River Salmon 2015 Season Summary and 2016 Season Outlook* (http://yukonriverpanel.com/salmon/publications/joint-technical-committee-reports/). Brief summaries of the outlook are presented below for each stock.

The comprehensive decision guidelines detailed below outline the management responses to a range of circumstances and the general rationale applied when making management decisions. Decision guidelines are meant to capture general management approaches that contribute to multi-year management planning.

For Yukon River fisheries in Canada, the management regime at the beginning of the season is guided by the pre-season run outlooks. Pre-season outlooks provide rough indications of run size and can be used to set expectations for the year (e.g., the run is unlikely to be large enough to support a commercial fishery in Canada). Pre-season decisions are made during the development of fishing plans and are based on pre-season run outlooks. These decisions are the result of technical analyses and are based on advice received during consultations and YRP meetings. Following consultations with Yukon First Nations, with commercial, domestic, and recreational fishery participants, and with members of the public, the YSSC provided recommendations to the Minister of Fisheries and Oceans Canada in June 2016. Their recommendations were accepted and incorporated into this management plan.

In-season assessment of Canadian-origin Yukon River Chinook and fall chum salmon occurs primarily in the lower Yukon River in Alaska (Pilot Station sonar and the Lower Yukon Test Fishery) and at the Eagle sonar assessment program located just downstream of the U.S./Canada border. As the season progresses, information from lower river assessments provides revised information on which management decisions may be made. This early-season information is often available in mid- to late-June and can provide improved certainty over pre-season information. This information serves to identify deviations from pre-season forecasts and allows time to prepare Canadian managers and fishers for potential changes to fishing plans. However, because of the difficulty in assessing salmon in the lower river, passage estimates of Canadian-origin salmon do not always have high certainty. In-season assessment information available from the Eagle sonar is very reliable and is the basis for making in-season management decisions in Canada.

Information sharing and dissemination

Fisheries and Oceans Canada will provide the YSSC and Yukon First Nation governments with periodic updates on the estimated abundance of Canadian-origin Chinook salmon returning to the Yukon River as well as in-season information on management measures and potential interception rates in Alaska. This will be accomplished through regular in-season (weekly) management teleconference hosted by DFO. The calls will provide the opportunity share information on the return of Canadian-origin salmon and provide the opportunity to discuss the management strategy and in-season changes.

8.1 Yukon River Mainstem Chinook Salmon Management

8.1.1 Pre-season Considerations and Decisions

The pre-season outlook is for a weak run, but a run that could support some level of fishing in Canada. The 2016 spawning escapement goal for Canadian-origin upper Yukon River Chinook salmon is of 42,500 to 55,000; this is the target number of Chinook salmon to reach the spawning grounds in the upper Yukon drainage in Canada. The anticipated pre-season outlook for Canadian-origin Chinook salmon is for a weak run of 65,000 to 88,000. Runs of 150,000 fish were common in 1980s and 1990s. Following a series of years where the lower end of the escapement goal has not been achieved (5 of the past 9 years), ensuring that a sufficient number of adult Chinook salmon reach spawning grounds in Canada is of paramount importance to sustaining the health of this stock into the future. In particular, the current period of low productivity and marginal stock recruitment rates (approximately only 1 adult produced for each spawning adult) indicates that a low rate of adult escapement is likely to preclude significant change in run sizes of Canadian-origin Chinook salmon in the near future.

Based on the pre-season outlook, the anticipated Total Allowable Catch for Canadian-origin Yukon River mainstem Chinook salmon is 10,000 to 45,500 (U.S. and Canada combined). Based on the mid-point of the catch share established by the Yukon River Salmon Agreement, Canada's catch share could be 3,700 to 9,000 Chinook salmon. At the low end of this run size range there would be an opportunity for some harvest in the First Nation subsistence fishery, but some conservation measures would be required. A return at the high end of the range would support a full (normal) First Nation subsistence fishery and potentially some openings in the recreational fishery.

Table 1. Canadian harvest shares and potential status of fisheries based on pre-season outlook range.

Outlook (Run Size)	Canadian Harvest Share*	Canadian Fisheries			
		First Nation	Recreational	Commercial	Domestic
65,000	3,700	3,700	Non-retention	Closed	Closed
76,500	6,400	6,400	Non-retention	Closed	Closed
88,000	9,000	8,000	Open	Closed	Closed

**The Canadian harvest share uses a mid-point management target of 48,750 and a 23% (mid-point) Canadian harvest share*

The considerations for the development of the 2016 management strategy were:

1. A pre-season outlook for another weak run (65,000 to 88,000) – similar in size to the 2015 run
2. In 5 of the past 9 years, the minimum spawning escapement target was not achieved leading to a concern about sustainability of the run and a number of potentially impacted brood years
3. The productivity of the run has been consistently low for the last 8 years

4. The 2015 run came in better than the pre-season forecast
5. There have been many years of sacrifice and conservation on behalf of fishers up and down the Yukon River, particularly in 2014 and 2015 where harvest of Chinook salmon were very low in the Yukon and Alaska
6. There are some ocean indicators of improved productivity
7. The productivity of the 2010 brood year (will lead to return of 6 year olds in 2016) will be at least as high as that of 2009 and maybe higher
8. There is an interest in some First Nations in having an opportunity to fish early in the run, based on traditional practices and knowledge
9. There is an interest in considering the 2016 management approach within a long-term management approach

To begin the season, the recreational, commercial, and domestic fisheries will be closed. An allocation will be made available to interested First Nations. However, in light of the considerations listed above, several recommendations around conservation measures will be included for early fishing opportunities before in-season assessment information provides greater certainty about run size. First, First Nations who do fish are requested to seek to reduce their harvest to 10% of a normal harvest. Second, where possible, harvest of Chinook salmon should be directed at smaller (younger) fish. This can be achieved through the use of smaller-mesh gill nets (i.e. 6" or less) or selective release or larger (older) fish from fish wheels or hook and line fisheries. Third, the sex ratio of returning salmon will be considered when making management decisions. The long-term historical ratio of males to females is approximately 55% males to 45% females. This beginning of season approach is intended to provide some protection for Chinook salmon early in the season until early- and in-season information can provide reliable projections of the run size. As confidence in in-season abundance improves, fishery management actions would proceed according to the *In-season fishery management decision matrix*.

This pre-season management approach will be modified based on early-season and in-season information. Early-season information (based on the Pilot Station sonar) will be used to modify and update these recommendations. If early season information does not suggest to a high degree of likelihood either a stronger or weaker run than anticipated then no changes to management will be made. In-season assessment of run strength (based on the Eagle sonar) will be used to inform management decisions in Canada.

8.1.2 In-Season Management

In-season fishery management decisions are based on the *In-season fishery management decision matrix for Yukon River Chinook salmon in Canada* (Table 2). In 2015, the matrix was updated to reflect the escapement goal range recommended by the Yukon River Panel. A mid-point management spawning escapement target ('Management Target') of 48,750 was adopted to improve the likelihood of achieving conservation objectives. These changes will carry forward in 2016. Further clarification of fisheries in the yellow zone has been provided. The matrix summarizes the management reference points, general allocation plans and anticipated management responses under different run size scenarios (as indicated by border passage):

RED ZONE < 42,500
No harvest – removal of all Chinook salmon harvest allocations. Run sizes this low represent a high conservation risk.

YELLOW ZONE (lower) 42,500 to 48,750
Base level (incidental) harvest in the First Nation fishery only. Anticipated harvest of less than 10% of annual subsistence needs.

YELLOW ZONE (upper) 48,750 to 55,000
Run supports some First Nation subsistence fishing. The harvest target is 10% to 90% of annual subsistence needs and varies in accordance with projected run abundance. Harvest targets are met using voluntary harvest reductions in each First Nation.

GREEN ZONE > 55,000
Opportunity for normal (full) First Nation subsistence harvest (i.e., no voluntary harvest reductions sought). Harvest opportunities (and allocation) for recreational, commercial, and domestic fisheries are provided in proportion to run abundance and are considered only when opportunities for First Nation harvests have not been restricted.

Table 2. In-season fishery management decision matrix for Yukon River mainstem Chinook salmon.

Border Passage Projections	Fishery			
	First Nation	Recreation	Commercial	Domestic
< 42,500	**Closed** Removal of allocation for conservation purposes.	**Closed** No retention permitted. Additional closures possible.	**Closed**	**Closed**
42,500 to 55,000 *Management Target: 48,750*[1]	**Varies** **42,500 to 48,750** Base level (incidental) harvest of less than 10% of annual subsistence needs. **48,750 to 55,000** Harvest target 10% to 90% of annual subsistence needs and varies with abundance	**Closed** No retention permitted	**Closed**	**Closed**
> 55,000	**Open** Unrestricted	**Open**[2] Retention permitted	**Open**[2] Allocation varies with run size	**Open**[2] Allocation varies with run size

¹ The Management Target of 48,750 is the minimum number of salmon intended to reach the spawning grounds.
² Allocations (harvest opportunities) are subject to run abundance and international harvest sharing provisions.

The border passage projection is the main indicator used to inform in-season management decisions, however harvest in Alaska before the fish reach Canada and the harvest shares established under the *Yukon River Salmon Agreement* are also considered when making management decisions.

8.2 Yukon River Mainstem Fall Chum Salmon Management

8.2.1 Pre-season Considerations and Decisions

The 2016 spawning escapement goal for Canadian-origin upper Yukon River chum salmon is of 70,000 to 104,000; this is the target number of chum salmon to reach the spawning grounds in the upper Yukon drainage in Canada. The anticipated pre-season outlook for Canadian-origin Chinook salmon is for a run of 137,000 to 195,000. Based on the pre-season outlook and aiming for the midpoint of the escapement goal range, the anticipated Total Allowable Catch for Canadian-origin Yukon River mainstem Chinook salmon is 50,000 to 108,500 (U.S. and Canada combined). Based on the mid-point of the catch share established by the Yukon River Salmon Agreement, Canada's catch share could be 16,000 to 34,600 chum salmon, suggesting opportunities will be available in all fisheries.

Table 3. Canadian harvest shares and potential status of fisheries based on pre-season outlook range.

Outlook (Run Size)	Canadian Harvest Share*	Canadian Fisheries			
		First Nation	Recreational	Commercial	Domestic
137,000	16,000	3,000	Open	12,900	100
166,000	25,300	3,000	Open	22,900	100
195,000	34,600	3,000	Open	31,500	100

**Assumes a spawning target of 87,000 and a 32% (mid-point) Canadian harvest share*

To begin the season, the recreational, commercial, domestic, and First Nation fisheries will be open. This pre-season management approach will be modified based on early-season and in-season information. Early-season information (based on the Pilot Station sonar) will be used to modify and update these recommendations. If early season information does not suggest to a high degree of likelihood either a stronger or weaker run than anticipated then no changes to management will be made. In-season assessment of run strength (based on the Eagle sonar) will be used to inform management decisions in Canada.

8.2.2 In-Season Management

In-season fishery management decisions are based on the in-season fishery management decision matrix for Yukon River chum salmon in Canada (Table 4). The current matrix is in need of revision as it does not align with the current spawning escapement goal range established by the Yukon River Panel. The Yukon Salmon Sub-Committee is planning consultations with First Nation governments, stakeholders, and the public to contemplate revisions. In the meantime, the matrix remains unchanged from 2015. The matrix summarizes the management reference points, general allocation plans and anticipated management responses under different run size scenarios (as indicated by border passage):

RED ZONE < 40,000
No harvest – removal of all chum salmon harvest allocations. Run sizes this low represent a high conservation risk.

YELLOW ZONE 40,000 to 73,000
Run supports some First Nation subsistence fishing. The harvest target varies in accordance with projected run abundance. Harvest targets are met using voluntary harvest reductions in each First Nation.

GREEN ZONE > 73,000
Opportunity for normal (full) First Nation subsistence harvest (i.e., no voluntary harvest reductions sought). Harvest opportunities (and allocation) for recreational, commercial, and domestic fisheries are provided in proportion to run abundance and are considered only when opportunities for First Nation harvests have not been restricted.

It should be noted that the border passage projections are the main indicators that inform management decisions are made in Canada and Fisheries and Oceans Canada also considers harvest shares established under the *Yukon River Salmon Agreement* when making management decisions; when salmon arrive at the border, Alaskan fisheries have already harvested a portion of the run.

Table 4. In-season fishery management decision matrix for Yukon River mainstem fall chum salmon.

Border Passage Projections	Fishery			
	First Nation	Recreation	Commercial	Domestic
< 40,000	**Closed** Removal of allocation for conservation purposes	**Closed** No retention permitted	**Closed**	**Closed**
40,000 to 73,000	**Varies**[1] Catch target to vary with abundance within zone	**Closed** No retention permitted	**Closed**	**Closed**
> 73,000	**Open** Unrestricted	**Open**[1] Retention permitted No catch anticipated	**Open**[1] Allocation varies with run size	**Open**[1] Allocation varies with run size

[1] Allocations (harvest opportunities) are subject to run abundance and international harvest sharing provisions.

The border passage projection is the main indicator used to inform in-season management decisions, however harvest in Alaska before the fish reach Canada and the harvest shares established under the *Yukon River Salmon Agreement* are also considered when making management decisions.

8.3 Porcupine River Chinook Salmon Management

8.3.1 Pre-season Considerations and Decisions

Currently, information limitations preclude the development of a formal forecast or outlook for Chinook salmon returning to the Porcupine River in Canada. In the absence of stock specific information, the general outlook for Porcupine Chinook salmon is based on an extension of the Mainstem Yukon River Chinook salmon outlook (i.e. a weak run with abundance similar in size to the 2015 return). Approximately 3,000 to 5,000 Chinook salmon are anticipated to return to the Porcupine River in 2016. Over the past 10 years the average Vuntut Gwitchin annual harvest has been approximately 275 Chinook salmon while the Basic Needs Allocation identified in the *Vuntut Gwitchin First Nation Final Agreement* is 750 Chinook salmon.

To begin the season, the First Nation fishery will be open with recommendations for a limited harvest. This pre-season management approach will be modified based on early-season and in-season information. Early-season information (based on the Pilot Station sonar) will be used to modify and update these recommendations. If early season information does not suggest to a high degree of likelihood either a stronger or weaker run than anticipated then no changes to management will be made. In-season assessment of run strength (based on the Old Crow Chinook salmon sonar) will be used to inform management decisions in Canada.

8.3.2 In-season Management

In-season fishery management decisions are based on information from the Old Crow Chinook salmon sonar. If in-season information suggests that restrictions are required within the First Nation fishery, DFO, the YSSC, and the Vuntut Gwitchin Government (VGG) will discuss potential conservation options in advance of implementation.

The Old Crow sonar passage projection is the main indicator used to inform in-season management decisions, however harvest in Alaska before the fish reach Canada is also considered when making management decisions.

8.4 Porcupine (Fishing Branch) River Fall Chum Salmon Management

8.4.1 Pre-season Considerations and Decisions

The 2016 spawning escapement goal for Porcupine (Fishing Branch) River fall chum salmon is 22,000 to 49,000; this is the target number of chum salmon to reach the spawning grounds on the Fishing Branch River. The pre-season outlook is for a run of 22,000 to 31,000. Harvest sharing provisions for Canadian-origin Porcupine River fall chum salmon stocks are not specified in the YRSA. Based on the pre-season outlook, the potential target harvest range in Canada is 0 to 2,900 fall chum salmon. Over the last 10 years, the average Vuntut Gwitchin harvest is approximately 3,000 fall chum salmon and the Basic Needs Allocation identified in the *Vuntut Gwitchin First Nation Final Agreement* is 6,000 fall chum salmon.

The considerations for the development of the 2016 management strategy were:

1. The escapement goal for Fishing Branch River chum salmon has not been achieved in 5 of the last 9 years.
2. Commercial and subsistence fisheries in Alaska harvest Fishing Branch River chum salmon in because they are mixed in with all other chum salmon migrating up the river.

Table 5. Canadian harvest targets and potential status of fisheries based on pre-season outlook range.

Outlook (Run Size)	Target Canadian Harvest*	Canadian Fisheries			
		First Nation	Recreational	Commercial	Domestic
22,000	0	Closed	n/a	n/a	n/a
27,000	1,600	1,600	n/a	n/a	n/a
31,000	2,900	2,900	n/a	n/a	n/a

**Assumes a spawning target of 22,000 and uses a 32% Canadian harvest share*

To begin the season, the First Nation fishery will be open with a recommendation to limit harvest to 30% or less of the recent subsistence harvest. This approach is intended to provide an opportunity for a modest level of subsistence harvest during the early part of the run until such time as a more robust in-season estimate may be derived from information collected through the Porcupine chum salmon sonar assessment program. As confidence in the run abundance is gained, fishery management actions would proceed accordance with abundance.

This pre-season management approach will be modified based on early-season and in-season information. Early-season information (based on Pilot Station sonar) will be used to modify and update these recommendations. If early season information does not suggest to a high degree of likelihood either a stronger or weaker run than anticipated then no changes to management will be made. In-season assessment of run strength (based on the Old Crow chum salmon sonar) will be used to inform management decisions in Canada.

8.4.2 In-season Management

In-season fishery management decisions are based on information from the Old Crow chum salmon sonar. Projection of the fall chum run size at Old Crow will be combined with estimates of the proportional return of Porcupine River chum salmon to the Fishing Branch River (approximately 60%) to develop a projection for return of fall chum salmon to the Fishing Branch River. This will be compared to the spawning escapement goal.

The Old Crow passage projection is the main indicator used to inform in-season management decisions, however harvest in Alaska before the fish reach Canada is also considered when making management decisions.

8.5 Selective Fisheries

Selective fishing is defined as the ability to avoid non-targeted fish (and could include specific sizes and/or sexes within species), invertebrates, birds, and mammals or, if encountered, to release them alive and unharmed. Selective fishing technology and practices will be adopted, where appropriate, in all fisheries in the Pacific Region, and there will be attempts to continually improve selective harvesting gear and related practices.

Selective harvesting standards will be set in the context of the *Policy for Selective Fishing in Canada's Pacific Fisheries* and the *Allocation Policy for Pacific Salmon*. In the future, priority will be given to those who have demonstrated the ability to meet or exceed the selective fishing standards. The Department encourages the incorporation of selective fishing experiments into regular fisheries where appropriate.

In the context of the Yukon River salmon fisheries, there is an interest in undertaking initiatives that use selective fishing gear which could allow for the release of larger-sized Chinook salmon, and the release of female Chinook salmon, or the ability to target smaller male fish. Traditional knowledge, anecdotal information, and recent scientific information, suggest that the average weight (and size) of Yukon River Chinook salmon has decreased over time. Two potential explanations for the decrease are historical fishing practices (which targeted larger-sized fish for many years) or variation in environmental conditions.

A selective fishery demonstration program using fish wheels received funding from the YRP in 2006, 2007 and 2008. In 2006, a fish wheel equipped with holding pens was operated during commercial fishery openings near a commercial fishing site to facilitate a comparison between the selective gear and the commercial catch in the same area during the same time period. The project demonstrated that fish wheels could catch as many fish as nets and allow successful release of females and larger fish. Unfortunately, few opportunities for commercial fisheries directed at Chinook salmon have occurred due to poor returns over the past several years.

8.6 By-catch Management

The *Allocation Policy for Pacific Salmon* describes priorities and considerations for the directed harvest of target stocks. However, these opportunities may have to be constrained due to conservation concerns for species, stocks or stock aggregates also encountered during these directed fisheries. The inadvertent harvest of different species of concern is referred to as by-catch. The inadvertent harvest of stocks of concern within the same species is referred to as incidental harvest. Both by-catch and incidental harvest are factored into the calculation of exploitation rates on various stocks, and therefore, fishing plans are designed to be consistent with existing policies and to keep exploitation rates on stocks of concern within the limits described in the conservation objectives.

Yukon River salmon migrate into the Bering Sea during the spring and summer after spending 0, 1, or 2 winters rearing in fresh water, depending on the species. Information on stock origin from tagging, scale pattern, parasites, and genetic analysis indicate that Yukon River salmon are present throughout the Bering Sea, in regions of the North Pacific Ocean south of the Aleutian chain, and the Gulf of Alaska during their ocean migration.

The by-catch of Chinook salmon in both the Alaskan Bering Sea - Aleutian Islands area (BSAI) and the Gulf of Alaska (GOA) groundfish fisheries rose significantly in the late 1990s and early 2000s, peaking in 2007. Since then, due to the implementation of a number of fishing strategies intended to minimize the interception of salmon, by-catch levels have been reduced considerably. For example, since 2012 it is estimated that fewer than 2,000 upper Yukon River Chinook salmon were incidentally intercepted in the Bering Sea groundfish fishery. Salmon by-catch in all Bering Sea and Gulf of Alaska groundfish fisheries is monitored through an on-board independent observer program. Except for donations to food banks, salmon cannot be retained or sold. In past years, concerns over the escalating by-catch have been expressed by the YRP to the North Pacific Fishery Management Council, which oversees the management of these groundfish fisheries. The primary concerns expressed was that interception of Yukon salmon in these fisheries is inconsistent with the YRSA, which obliges the Parties to decrease the marine catch and by-catch of Yukon River salmon. Although the situation has improved considerably over recent years, ongoing monitoring and reporting on salmon by-catch in BASI and GOA groundfish fisheries is imperative in ensuring that interception of Yukon River stocks continues to be minimized.

9. SHARED STEWARDSHIP ARRANGMENTS

Stewardship refers to the care, supervision or management of something, especially the careful and responsible management of something entrusted to one's care.[4] In the context of fisheries management, stewardship is often considered in terms of "shared stewardship", whereby First Nations, fishery participants and other interests are effectively involved in fisheries management decision-making processes at appropriate levels, contributing specialized knowledge and experience, and sharing in accountability for outcomes.

Moving toward shared stewardship is a strategic priority for DFO. This is reflected in a number of policies and initiatives, including the WSP, the Resource Management Sustainable Fisheries Framework (SFF), Fisheries Reform, Aboriginal Aquatic Resource and Oceans Management (AAROM) Program and the AFS.

Also referred to as "co-management," DFO is advancing shared stewardship by promoting collaboration, participatory decision making and shared responsibility and accountability with resource users and others. Essentially, shared stewardship means that those involved in fisheries management work cooperatively—in inclusive, transparent and stable processes—to achieve conservation and management goals.

Consultation and engagement with First Nations is central to DFO's approach to fisheries management (including the development of management strategies described within this IFMP) and fulfilling the Department's mandate. In addition to supporting good governance, sound policy and effective decision-making, Canada has statutory, contractual and common-law obligations to consult with Aboriginal groups. For example, The Crown has a legal duty to consult and, if appropriate, accommodate, when the Crown contemplates conduct that might adversely impact *Fisheries Act Section 35* rights (established or potential).

Consultation and engagement with First Nations takes place at a number of levels and through a variety of processes. For example, a significant amount of consultation and dialogue takes place through direct, bilateral meetings between DFO and First Nations at a local level. This can include specific engagement during the pre-season, in-season or post-season planning and reporting processes.

9.1 Consultative Processes

The development of decision guidelines and specific management measures involves consultation with various First Nation government representatives, groups, individuals as well as coordinated efforts through the SSC. In the Yukon, consultative processes have been established for some time, particularly through implementation of First Nation Final Agreements. International consultation

[4] As defined in the Atlantic Fisheries Policy Review (AFPR)

has been established through the YRSA and the YRP. The consultative processes for these and other initiatives are described below.

9.1.1 Yukon Salmon Sub-Committee

The YSSC is a public advisory body established under Chapter 16 of the UFA. The mandate of the YSSC is to provide for the public input into matters related to salmon through their authority to make official recommendations to the Minister of Fisheries and Oceans and Yukon First Nations. These recommendations, although focusing on salmon harvest management, may pertain to legislation, research, policies, and programs.

The members of the YSSC come from all parts of the Yukon and are comprised of both First Nation and non-First Nation people. The composition of the ten-member Committee is laid out in the UFA and is carefully structured to ensure diversity and balance. YSSC members consist of Yukon Fish and Wildlife Board appointees, nominees from Fisheries and Oceans Canada and the First Nations of the Alsek, Porcupine, and Yukon River drainage basins.

The YSSC established the Yukon River and Porcupine River working groups to gather input and comment on matters related to salmon. Participants in the working groups may include First Nation representatives, commercial, domestic and recreational fishers, Renewable Resource Council members, selected members of the YSSC and technical staff from Fisheries and Oceans Canada. Additionally, community and public meetings are convened annually by the YSSC to discuss run outlooks, present an allocation framework and decision matrix, and receive public input. The results of these efforts is an official recommendation from the YSSC to the Minister of Fisheries and Oceans Canada on the allocation of total allowable catch as outlined in *s 16.10.1, 16.10.2* and *s 16.7.17.12 (f)* of the UFA.

9.1.2 Yukon River Panel

The YRP was established pursuant to the YRSA and consists of six Canadian members and six United States members. Each party is responsible for the appointment of its members. As per section *16.7.17.13* of Yukon First Nation Final Agreements, YSSC members must constitute the majority of the Canadian section of the YRP.

To implement the YRSA, the YRP may make recommendations to the management agencies regarding various topics including: conservation, restoration, rebuilding and management of salmon stocks originating in the Yukon River in Canada; the coordination of management plans and actions of the Yukon River fisheries that affect Canadian-origin salmon stocks; and projects to be funded under the REF.

The YRP is supported by its Joint Technical Committee (the JTC) which prepares and reviews post-season summaries, stock status, escapement goal analyses, research project reports including

Chinook size trend analyses, research planning initiatives, and restoration and enhancement proposals.

9.1.3 First Nation Aboriginal Fisheries Strategy Consultations

Engagement relating to the AFS occur throughout the year with Yukon First Nations that have not yet concluded Final Agreements. These meetings help to inform the content of DFO/First Nation Fisheries Agreements. The Agreements may contain details pertaining to fisheries activities and programs occurring in First Nation traditional territories such as communications, management of First Nation fisheries, stock assessment, habitat and enhancement programs, enforcement protocols and communal licences. The AFS agreement documents, as well as records of consultation sessions and progress on action items, are maintained by the AFS Coordinator.

10. COMPLIANCE PLAN

10.1 Conservation and Protection Program Description

The Conservation and Protection (C&P) program promotes and maintains compliance with legislation, regulations and management measures implemented to achieve the conservation and sustainable use of Canada's aquatic resources, and the protection of species at risk, fish habitat and oceans.

The program is delivered through a balanced regulatory management and enforcement approach including:

- promotion of compliance through education and shared stewardship;
- monitoring, control and surveillance activities;
- Management of major cases /special investigations in relation to complex compliance issues.

In carrying out activities associated with the management of Pacific salmon as outlined in this management plan, C&P will utilize principle-based approaches and practices which are consistent with the National Compliance Framework and the DFO Compliance Model.

10.2 Compliance Program Delivery

For the salmon fisheries in the Pacific Region, C&P will be utilizing a broad scope of tools and approaches to manage compliance towards achieving conservation and sustainability objectives, including:

- Maintain and develop relationships with First Nations communities, recreational groups and commercial interests through dialogue, education and shared stewardship.
- Intelligence-led investigations may specifically target repeat and more serious offenders for increased effectiveness of enforcement effort. Illegal sales of salmon will continue to be a regional priority.
- Prioritize enforcement efforts on measures directed towards conservation objectives.
- Fish habitat protection remains a key focus of fishery officer efforts coordinated regionally by the Fisheries Protection Program.
- Utilize 'Integrated Risk Management' to ensure fishery officer efforts are focused and directed at problems of highest risk.
- Continue high profile fishery officer presence through patrols by vehicle, vessel and aircraft to detect and deter violators.

- Support traceability initiatives within the salmon fishery to enhance accountability. Monitor and verify catches and offloads of salmon to ensure accurate and timely catch reporting and accounting, including coverage of Dual Fishing opportunities.
- Priorities and direct compliance efforts where there is a risk to salmon stocks of concern.
- Use of enhanced surveillance techniques, and new available technology as well as covert surveillance techniques as a means to detect violations and gather evidence in fisheries of concern.
- Patrols during open timed fisheries to increase intelligence gathering, build relationships with stake holders and ensure compliance to licence conditions.
- Inspect fish processors, cold storage facilities, restaurants and retail outlets for compliant product.
- Maintain a violation reporting 24-hour hotline to facilitate the reporting of violations.
- Continue to promote 'Restorative Justice' principles in all fisheries.

10.3 Consultation

C&P works closely within the Fisheries and Aquaculture Management sector to ensure that fishery management plans are enforceable and implemented in a controlled, fair manner. C&P has a multi-faceted role as educator, referee, mediator and law enforcer.

C&P participates on a regular basis in consultations with the fishing community and general public. Education, information and shared stewardship are a foundation of C&P efforts. C&P participates in all levels of the advisory process. The importance of local field level fishery officer input to these programs has proven invaluable and will continue.

C&P will continue meeting at the local level with individual First Nations, through the fishery officer First Nation Liaison Program and with First Nations planning committee meetings that involve many First Nations' communities at one time.

Fishery officers are viewed as the public face of the department. During their day-to-day activities, the fishing community and general public provide comment and input that is promptly communicated to C&P managers, fisheries managers and Fisheries Protection Program staff. This public feedback is critical in identifying issues of concern and providing accurate feedback on emerging issues.

10.4 Compliance Strategy

In 2016, specific objectives for the salmon fishery will be to focus compliance management efforts on:

- Maintaining enhanced coverage both on in-river and land based approaches by undertaking vessel, vehicle, and air patrols;
- Curtailing illegal sales through a program designed to improve traceability of catch (improved catch monitoring and plant / storage verification);
- Balancing close time patrols with random open time patrols;
- Working with stakeholders to improve regulatory compliance;
- Enforcing daily limits, non-retention and compliance with closed area restrictions in the recreational fishery;
- Maintaining or increasing efforts to protect Yukon River salmon stocks with priority to those stocks of concern such as Chinook salmon;
- Increasing Fishery Officer efforts to protect other Yukon River salmon stocks of concern through implementation of area-specific project management enforcement plans;
- Monitoring and auditing catches, effort and offloads/landings of salmon during fisheries to ensure timely and accurate catch and effort reporting.

The management of Pacific salmon and salmon fishery compliance continues to be a high priority for 2016. In order to balance multiple program demands, C&P applies a risk-based integrated work planning process at the Regional and Area levels to establish annual operational priorities that address the highest risks to sustainability. This process ensures that resources are allocated in alignment with identified priorities to achieve broad departmental objectives in a way that best serves the interests of Canadians. Resource utilization is dependent on availability of program funding. C&P cannot be effective without the commitment of all salmon fishers and the fishing industry to the conservation of this valuable resource.

11. APPENDIX 1: 2015 SEASON REVIEW AND STOCK STATUS

11.1 2015 Season Review

At the conclusion of each season, the Yukon River Panel's Joint Technical Committee meet to review the fishing season and prepare an annual report which contains a description and results of fishing activities, management strategies, total run size, historical catch, and escapement information for Canadian-origin salmon stocks designated by the Yukon River Salmon Agreement (YRSA). The "*Yukon River Salmon 2015 Season Review and 2016 Outlook*" is available through the Yukon River Panel's website:
http://yukonriverpanel.com/salmon/wp-content/uploads/2009/03/rir3a201601.pdf.

Performance of 2016 fisheries and management measures in both Canada and the U.S. are assessed based on harvest data submitted by subsistence, recreational, commercial and domestic fisheries. For the Canadian portion of the Yukon River watershed, recreational catch information is collected via the Yukon Salmon Conservation Catch Card while First Nation fisheries provide harvest monitoring data through monitoring completed by individual First Nations. Domestic and commercial catches are evaluated through mandatory harvest reporting.

Appendix D of the U.S. / Canada Joint Technical Committee (JTC) report contains a 'report card' that outlines escapement goals, run size, allowable and actual catches in the U.S. and Canada, and the spawning escapement for Canadian-origin Chinook and chum salmon. This report is the best sources of data on the 2015 (and previous) Yukon River salmon runs.

11.2 Historic and Current Stock Status - Chinook Salmon

Through the 1980s, Chinook escapements in the Canadian section of the Yukon River drainage were in a state of decline. A plan to prevent further declines, while formulating rebuilding plans, was developed jointly in the Canada/U.S. Yukon River salmon negotiations and adopted as part of the *Interim Yukon River Salmon Agreement* (IYRSA), which was signed in February 1995. In this plan, a stabilization spawning escapement goal of 18,000 Canadian-origin upper Yukon Chinook was established for the period 1990 through 1995. A target escapement goal range of 33,000 to 43,000 Chinook salmon was also agreed to for rebuilt runs.

In accordance with the IYRSA, the parties were tasked with developing a Chinook rebuilding plan and providing recommendations regarding the implementation of such a plan after the 1995 season. In April 1996, the Yukon River Panel agreed to implement an upper Yukon Chinook rebuilding plan by establishing a revised interim minimum escapement target of >28,000 Chinook salmon for 1996 through 2002. In 2003, the escapement target was 25,000 Chinook salmon, but was to be increased to 28,000 in the event a U.S. commercial fishery was initiated. In 2004, the escapement target for Canadian-origin upper Yukon Chinook salmon was >28,000 Chinook salmon. If the run was sufficiently strong, the escapement target could range up to 38,000 Chinook salmon, although the

Panel did not describe what constituted a "strong run". In 2005 and 2006, the escapement target for Canadian-origin upper Yukon Chinook salmon remained unchanged at >28,000 Chinook salmon as the run was not yet considered to be rebuilt. The arrangement for 2007 was consistent with the *Yukon River Salmon Agreement*. Since the 2007 run was deemed to be rebuilt (since the primary brood year escapements achieved the escapement goal range for rebuilt stocks), the long-term escapement target of 33,000 to 43,000 was in effect. In 2008 and 2009, the escapement goal was changed to >45,000 upper Yukon Chinook salmon to make it consistent with the new method of assessing border escapement and spawning escapement (i.e. sonar). In 2010, an escapement range was established (42,500 to 55,000 Canadian-origin Chinook salmon) and this range has been renewed annually by the Yukon River Panel.

11.3 Historic and Current Stock Status - Fall Chum Salmon

As with Yukon River Chinook salmon, fall chum salmon are the target of numerous fisheries located throughout the river and in approach areas in marine waters. For example, in addition to the U.S. in-river harvest, significant catches of Yukon-origin fall Chum salmon are believed to occur in U.S. fisheries along the Aleutian Islands chain in some years, particularly near False Pass. Throughout the 1980s, Canadian fall chum escapements appeared to be depressed. In the YRSA, Canada and the U.S. agreed to rebuild the fall chum spawning escapements to more than 80,000 fish in the upper Yukon and to the 50,000-120,000 range in the Fishing Branch River.

12. APPENDIX 2: LANDINGS AND MARKETS

12.1 Landings

In recent years, Canadian harvest of Yukon River Chinook salmon has largely been driven by annual run abundance. Since 2007, the primary harvest of Chinook salmon has occurred through Aboriginal subsistence fisheries, although in most years the extent of these fisheries has been limited due to concerns over achieving sufficient spawning escapement. Canadian harvest of Chinook salmon in Yukon River fisheries since 1993 is summarized in Appendix Table 1.

Appendix Table 1. Canadian harvest of Yukon River Chinook salmon: 1993 to 2015.

Year	Yukon River Mainstem						Porcupine River	Canadian
	Aboriginal	Commercial	Recreational	Domestic	Test Fishery	Total	Aboriginal	Total
1993	5,576	10,350	300	243		16,469	142	16,611
1994	8,069	12,028	300	373		20,770	428	21,198
1995	7,942	11,146	700	300		20,088	796	20,884
1996	8,451	10,164	790	141		19,546	66	19,612
1997	8,888	5,311	1,230	288		15,717	811	16,528
1998	4,687	390	Closed	24	737	5,838	99	5,937
1999	8,804	3,160	177	213		12,354	114	12,468
2000	4,068	Closed	Closed	Closed	761	4,829	50	4,879
2001	7,421	1,351	146	89	767	9,774	370	10,144
2002	7,139	708	128	59	1,036	9,070	188	9,258
2003	6,121	2,672	275	115	263	9,446	173	9,619
2004	6,483	3,785	423	88	167	10,946	292	11,238
2005	6,376	4,066	436	99		10,977	394	11,371
2006	5,757	2,332	606	63		8,758	314	9,072
2007	4,175	Closed	Closed	Closed	617	4,794*	300	5,094
2008	2,885	Closed	Closed	Closed	513	3,399*	314	3,713
2009	3,791	364	125	17		4,297	461	4,758
2010	2,455	Closed	Closed	Closed		2,456*	250	2,706
2011	4,550	Closed	40	Closed		4,594*	290	4,884
2012	2,000	Closed	Closed	Closed		2,000	200	2,200
2013	1,902	Closed	Closed	Closed		1,904*	242	2,146
2014	100[a]	Closed	Closed	Closed		100	3	103
2015	1,000	Closed	Closed	Closed		1,000	204	1,204

* Totals include any incidental harvest. [a] Data are preliminary.

Harvest of fall chum salmon in Canadian Yukon River fisheries since 1993 are summarized in Appendix Table 2. Although abundance and resulting harvest levels have fluctuated in recent years, market conditions have had the greatest effect on commercial harvest.

Appendix Table 2. Canadian harvest of Yukon River fall chum salmon: 1993 to 2015.

Year	Yukon River Mainstem				Porcupine River	Canadian Total
	Aboriginal	Commercial	Domestic	Total	Aboriginal	
1993	4,660	7,762	0	12,422	1,668	14,090
1993	5,319	30,035	0	35,354	2,654	38,008
1995	1,099	39,012	0	40,111	5,489	45,600
1996	1,260	20,069	0	21,329	3,025	24,354
1997	1,238	8,068	0	9,306	6,294	15,600
1998	1,795	Closed	Closed	1,795	6,159	7,954
1999	3,234	10,402	0	13,636	6,000	19,636
2000	2,927	1,319	0	4,246	5,000	9,246
2001	3,077	2,198	3	5,278	4,594	9,872
2002	3,167	3,065	0	6,232	1,860	8,092
2003	1,493	9,030	0	10,523	382	10,905
2004	2,180	7,365	0	9,545	205	9,750
2005	2,035	11,931	13	13,979	4,593	18,572
2006	2,521	4,096	0	6,617	5,179	11,796
2007	2,221	7,109	0	9,330	4,500	13,830
2008	2,068	4,062	0	6,130	3,436	9,566
2009	820	293	0	1,113	898	2,011
2010	1,523	2,186	0	3,709	2,078	5,787
2011	1,000	5,312	0	6,312	1,851	8,163
2012	700	3,205	0	3,905	3,118	7,023
2013	500	3,369	18	3,887	2,283	6,170
2014	546[a]	2,485[a]	19	3,050	1,983[a]	5,033
2015	1,000	2,862	35	3,897	556	4,453

[a] Data are preliminary.

Historically the majority of harvest of Canadian-origin Yukon River salmon occurs in Alaskan subsistence and commercial fisheries. For Chinook salmon, ADF&G estimates the catch of Canadian-origin Chinook salmon by Alaskan fishers through genetic stock analyses. On average, approximately 50% of Chinook salmon harvested in the lower Yukon River (U.S. waters) were Canadian-origin fish.

12.2 Markets for Commercial Fish

The Han Fish Plant in Dawson City began operation in 1981 and became the largest Canadian market for commercially caught Yukon salmon. Products included fresh/frozen Chinook and fall Chum salmon as well as roe. Some experimentation with smoked products also occurred however due to a number of factors (primarily lack of sustained harvest opportunities for commercial fisheries) the plant has not operated since 1996. The lack of a major fish processor in Dawson City has had an overall negative effect on the commercial viability of the fishery and this partially explains the decrease in commercial harvests of Chinook and fall chum salmon since 1997 (Appendix Tables 1 and 2).

13. APPENDIX 3: FISHERY PLANS

Plans for each fishery will be based on the in-season fishery management decision matrices (see section 9) and in-season assessments of run strength.

13.1 First Nation Fishery Plan

Once conservation (spawning) requirements are achieved, constitutionally protected First Nation subsistence fisheries (for food, social, and ceremonial purposes) are provided priority allocation in management processes. Prior to reducing or removing the total allowable catch allocation available for First Nation subsistence fisheries, all other targeted salmon fisheries (commercial, recreational, and domestic) will be closed or have catch limits varied to zero.

Chinook salmon – some opportunity for First Nation harvest of Chinook salmon is anticipated in 2016. If the run comes in towards the low end of the pre-season forecast range, conservation measure will be required. If the run comes in near the high end of the pre-season forecast range, an opportunity for a full (normal) subsistence fishery will be available. To begin the season, the Yukon Salmon Sub Committee recommended a 10% of normal harvest until in-season information is available to better assess the run size. To ensure that recommendations about harvest level in the First Nation subsistence fisheries are implemented only to the extent reasonably necessary to achieve the conservation objective, DFO will closely monitor the return of Chinook salmon into Canada via in-river assessment programs to determine a run projection as early as possible.

For mainstem Canadian-origin fall chum salmon, opportunities for full First Nation subsistence fisheries are anticipated.

To promote the participation of First Nations in the administration and management of fisheries within respective traditional territories, Fisheries and Oceans Canada will continue to develop co-management initiatives with "Non Final Treaty" Yukon First Nations through the Department's AFS program. One of the primary avenues for accomplishing this is through the negotiation of annual Project Funding Agreements. In addition, management objectives are achieved through the development of Enforcement Protocols and the issuance of Communal Fishing Licences. Although communal licences are developed to fit the particular circumstances of each First Nation and reflect Final Agreement provisions, where they are in place, they all must be consistent with the principles set out constitutionally and associated Supreme Court rulings.

13.2 Recreational Fishery Plan

Chinook salmon – Due to the poor pre-season forecast for Yukon River Chinook salmon, no recreational harvest opportunities are anticipated in 2016 unless the run materializes stronger than anticipated. Effective mid-June, the permitted retention of Chinook salmon will be varied to 0.

Area-specific or regional angling closures may also be implemented to provide protection for spawning salmon during sensitive periods.

Chum salmon – Although an abundance of fall chum salmon is anticipated, participation in this fishery has been limited due to timing (late in the year) and difficulty in accessing fishing areas.

DFO fishery officers will conduct in-season monitoring of the recreational fisheries through enforcement patrols. All anglers fishing for salmon must have a valid Yukon Angling Licence as well as a Salmon Catch Card. A Regulation Summary is provided out when licences are issued and includes information on area-specific closures, hook sizes and other fishing gear limitations that must be adhered to when fishing for salmon. Anglers are advised to consult the *Yukon Territory Fishery Regulations* for further details *(http://lois.justice.gc.ca/en/showtdm/cr/C.R.C.-c.854)*. All recreational salmon anglers are reminded that the completed Salmon Catch Card must be returned to Fisheries and Oceans Canada by November 30.

13.3 Domestic Fishery Plan

Chinook salmon – No domestic fishery opportunities for Chinook salmon are anticipated in 2016.

Chum salmon – An anticipated abundance of mainstem fall chum salmon is likely to provide for harvest opportunities for domestic fisheries. Openings will likely be planned to coincide with commercial fishery openings. When openings are provided, fishers will be required to report catches, tag recovery and associated data within eight hours after the closure of each fishery. Information can be mailed in to the Fisheries and Oceans Canada office in Whitehorse, or telephoned to the toll free number: 1-877-salmon2 (1-877-725-6662). Further information on reporting requirements can be received by contacting Fisheries and Oceans Canada. DFO fishery officers will conduct monitoring of the domestic fishery.

13.4 Commercial Fishery Plan

Chinook salmon – No directed commercial harvest opportunities for Yukon River Chinook salmon are anticipated in 2016.

Chum salmon – An abundance of mainstem chum salmon is anticipated to provide harvest opportunities for commercial fisheries. Weekly fishing times will be based on in-season assessments of run strength and run timing, escapement projections and the status of the cumulative catch relative to the Total Allowable Catch. Therefore, week-to-week adjustments in fishing time should be expected. DFO will endeavor to announce weekly fishing times 48 hours prior to the proposed opening date however announcements of extensions and emergency closures may be made on shorter notice.

Announcements of openings/closures will be made via Fisheries and Oceans Canada's fishery notification system (http://www-ops2.pac.dfo-mpo.gc.ca/fns-sap/index-eng.cfm).

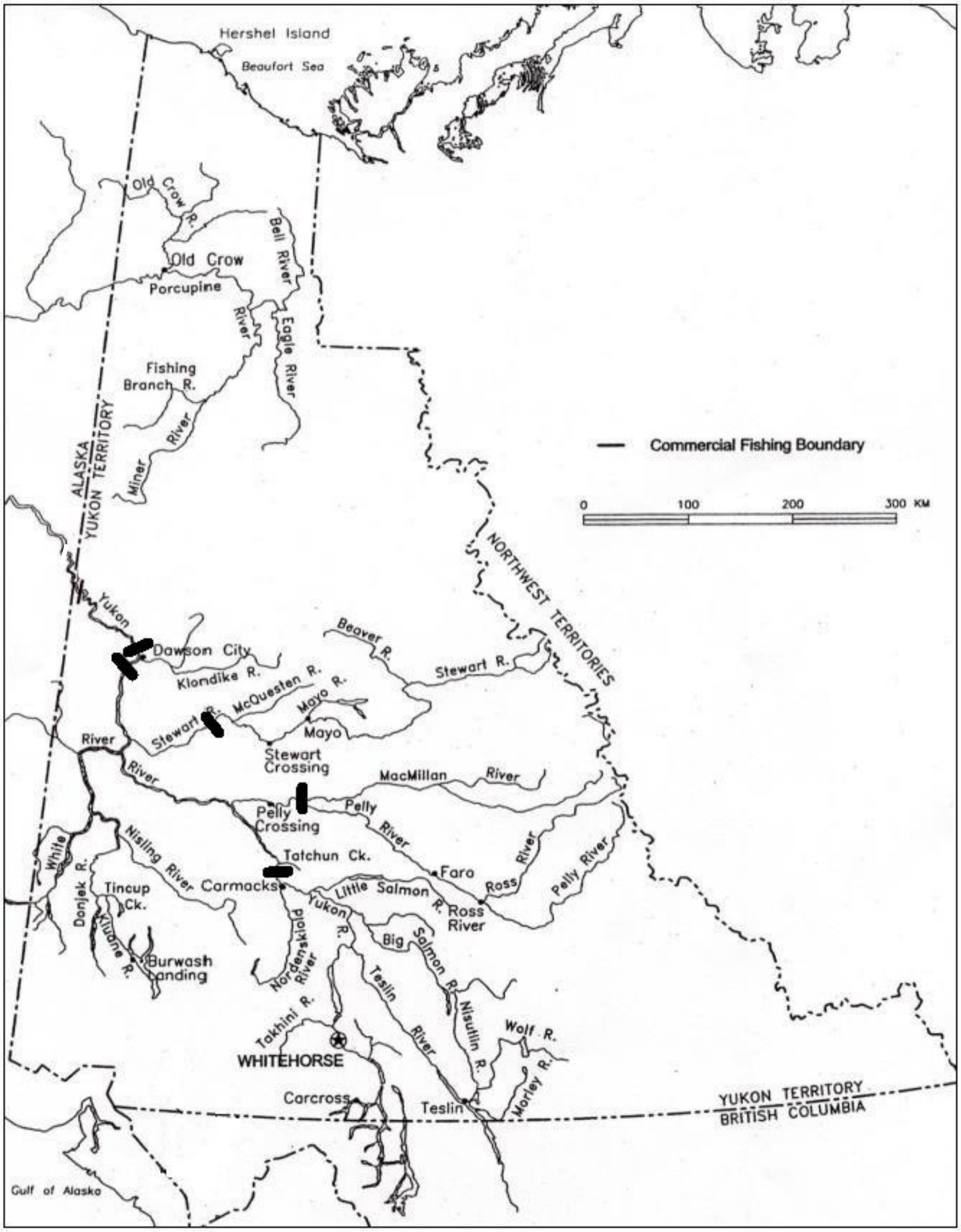
DFO fishery officers and stock assessment personnel will conduct monitoring of commercial fishing activities. There is an increasing responsibility for commercial fishers to accurately document catches and report this information weekly. These responsibilities will be clearly outlined in conditions attached to commercial licences. Gear allowances will remain unchanged from previous years and as described in the *Yukon Territory Fishery Regulations*.

Commercial fishers must keep records of daily catch and tag recovery data, tabulated on forms provided by Fisheries and Oceans Canada. Within eight hours of the closure of each weekly fishing period, fishers will be required to report this information by:

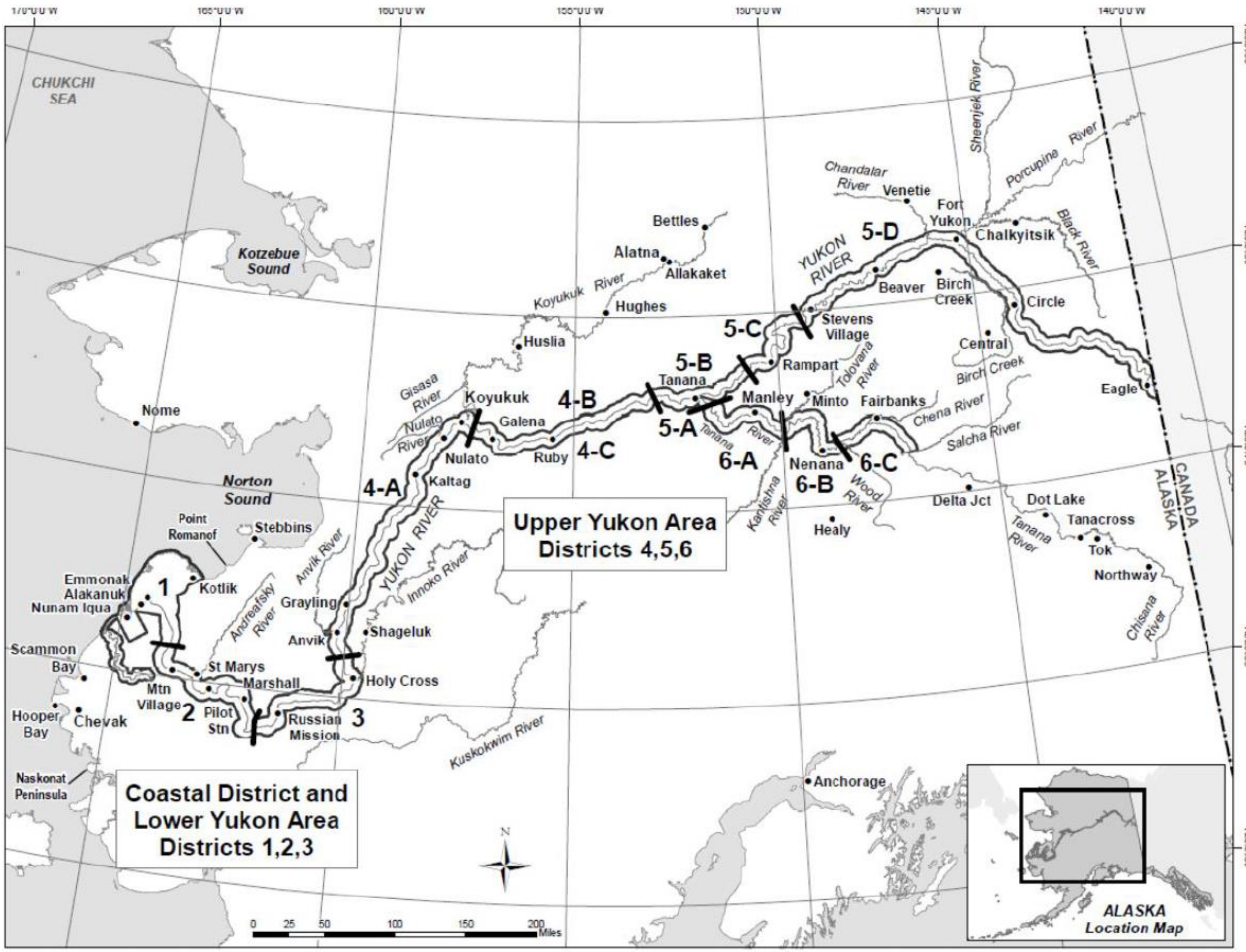
a. Phoning in their daily catch and tag recovery information to the Fisheries and Oceans Canada toll free catch line: 1-877-salmon2 (1-877-725-6662). Phoned-in information shall include:
    - The name of the fisher;
    - The catch of each species per day broken down by numbers of males and females;
    - The number of tags recovered from each species each day broken down by tag colour.

b. And, fishers are also required to mail their information no later than 10 business days after a weekly fishery closure to: Fisheries and Oceans Canada, Suite 100 – 419 Range Road, Whitehorse, Yukon. Y1A 3V1.

14. APPENDIX 4: MAPS OF FISHING AREAS



Appendix Figure 1. Yukon River drainage in Canada (dark bars delineate commercial fishing zones).



Appendix Figure 2. Yukon River drainage in Alaska (dark bars and numbers delineate fishing districts).

15. APPENDIX 5: LEGISLATION

The following Acts, Regulations and Agreements inform the management of Yukon River salmon:

- *Fisheries Act*
- *Pacific Salmon Treaty – Yukon River Salmon Agreement*
- Yukon First Nation Final and Self-Government Agreements
- *Yukon Territory Fishery Regulations* (pursuant to *Fisheries Act*)
- *Fishery (General) Regulations*
- *Aboriginal Communal Fishing Licence Regulations*
- *Management of Contaminated Fisheries Regulations*

16. APPENDIX 6: GLOSSARY

Aboriginal Traditional Knowledge (ATK) or Traditional Ecological Knowledge (TEK): Knowledge that is held by, and unique to Aboriginal peoples. It is a living body of knowledge that is cumulative and dynamic and adapted over time to reflect changes in the social, economic, environmental, spiritual and political spheres of the Aboriginal knowledge holders. It often includes knowledge about the land and its resources, spiritual beliefs, language, mythology, culture, laws, customs and medicines.

Abundance: Number of individuals in a stock or a population.

Age Composition: Proportion of individuals of different ages in a stock or in the catches.

Anadromous: An anadromous species, such as salmon, spends most of its life at sea but returns to fresh water grounds to spawn in the river it comes from.

Border Passage: The number of adult, upstream migrating salmon that escape all U.S. fisheries and reach the Canada/U.S. border.

By-catch: The unintentional catch of one species when the target is another.

Catch per Unit Effort (CPUE): The amount caught for a given fishing effort. Ex: tons of shrimp per tow, kilograms of fish per hundred longline hooks.

Coded Wire Tag (CWT): A small metal tag inserted into the nose of a juvenile salmon (usually hatchery stock) prior to release or migration to the ocean. The tag has encoded information that indicates the origin and year of release of the fish

Communal Commercial Licence: Licence issued to Aboriginal organizations pursuant to the *Aboriginal Communal Fishing Licences Regulations* for participation in the general commercial fishery.

Committee on the Status of Endangered Wildlife in Canada (COSEWIC): Committee of experts that assess and designate which wild species are in some danger of disappearing from Canada.

Discards: Portion of a catch thrown back into the water after they are caught in fishing gear.

Ecosystem-Based Management: Taking into account of species interactions and the interdependencies between species and their habitats when making resource management decisions.

Escapement: Reference to salmon - the number of fish escaping the fishery and reaching the spawning grounds.

Fishing Effort: Quantity of effort using a given fishing gear over a given period of time.

Fixed Gear: A type of fishing gear that is set in a stationary position. These include traps, weirs, gillnets, longlines and handlines.

Food, Social and Ceremonial (FSC): A fishery conducted by Aboriginal groups for food, social and ceremonial purposes.

Gillnet: Fishing gear: netting with weights on the bottom and floats at the top used to catch fish. Gillnets can be set at different depths and are anchored to the seabed.

Mark-Recapture Program: A stock assessment program that has a primary objective of estimating the size of populations. It usually involves live-capturing fish, marking or tagging them and releasing them back into the water at one location. At a second location, attempts are made to recapture both tagged and untagged fish. Tag and recapture data are used to generate population estimates

Maximum Sustainable Yield (MSY): Largest average catch that can continuously be taken from a stock.

Mesh Size: Size of the mesh of a net. Different fisheries have different minimum mesh size regulation.

Otolith: Structure of the inner ear of fish, made of calcium carbonate. Also called "ear bone" or "ear stone". Otoliths are used to determine the age of fish: annual rings can be observed and counted. Daily increments are visible as well on larval otoliths.

Population: Group of individuals of the same species, forming a breeding unit, and sharing a habitat.

Precautionary Approach: Set of agreed cost-effective measures and actions, including future courses of action, which ensures prudent foresight, reduces or avoids risk to the resource, the environment, and the people, to the extent possible, taking explicitly into account existing uncertainties and the potential consequences of being wrong.

Quota: Portion of the total allowable catch that a unit such as vessel class, country, etc. is permitted to take from a stock in a given period of time.

Recruitment: Amount of individuals becoming part of the exploitable stock (e.g., that can be caught in a fishery).

Research Survey: Survey at sea, on a research vessel, allowing scientists to obtain information on the abundance and distribution of various species and/or collect oceanographic data. For example may be a bottom trawl survey, plankton survey, or hydroacoustic survey.

Return per spawner: the average number of salmon produced from one spawning salmon

SARA: The *Species at Risk Act* is a federal government commitment to prevent wildlife species from becoming extinct and secure the necessary actions for their recovery. It provides the legal protection of wildlife species and the conservation of their biological diversity.

Scale Pattern Analysis: different freshwater rearing conditions may be reflected in different growth rates that can create varying/unique scale patterns that allow general point of origin assessments to be made

Spawner: Sexually mature individual.

Spawning Escapement: The number of adult salmon that escape all fisheries and return to the spawning grounds

Spawning Stock: Sexually mature individuals in a stock.

Stock: Describes a population of individuals of one species found in a particular area, and is used as a unit for fisheries management. Ex: NAFO area 4R herring.

Stock Assessment: Scientific evaluation of the status of a species belonging to a same stock within a particular area in a given time period.

Subsistence Fishery: A fishery that fills a need for food purposes. In Canada, it is not to be confused with the aboriginal fishery, which is restricted to First Nation members. In Alaska, the subsistence fishery involves both aboriginal and non-aboriginal Alaskan residents

Total Allowable Catch (TAC): The amount of catch that may be taken from a stock or the amount of fish that can be harvested after accounting for a specified spawning escapement target.

Traditional Ecological Knowledge (TEK): A cumulative body of knowledge and beliefs handed down through generations by cultural transmission, about the relationship of living beings (including humans) with one another and with their environment.

Tonne: Metric tonne, which is 1000kg or 2204.6lbs.

Year-class: Individuals of a same stock born in a particular year. Also called "cohort".

17. APPENDIX 7: ACRONYMS

ABM	Abundance-Based Management
ADF&G	Alaska Department of Fish and Game
AFS	Aboriginal Fisheries Strategy
ATP	Allocation Transfer Program
BASIS	Bering-Aleutian Salmon International Survey
BNA	Basic Needs Allocations
C&P	Conservation and Protection unit
COSEWIC	Committee On the Status of Endangered Wildlife In Canada
CWT	Coded-wire tag.
CU	Conservation Unit as identified in the Wild Salmon Policy
CYFN	Council for Yukon First Nations
DFO	Fisheries and Oceans Canada
FN	First Nation
FSC	Food, social and ceremonial
GSI	Genetic Stock Identification
IFMP	Integrated Fisheries Management Plan
IMEG	Interim Management Escapement Goal
IYRSA	*Interim Yukon River Salmon Agreement*
JTC	Joint Technical Committee of the Yukon River U.S./Canada Panel
MRP	Mark-Recapture Program
NMFS	National Marine Fisheries Service (U.S.)
PSARC	Pacific Scientific Advice Review Committee
PSC	Pacific Salmon Commission
PST	Pacific Salmon Treaty
R/S	Return per spawner
RCMP	Royal Canadian Mounted Police
RRC	Renewable Resources Council
SARA	Species at Risk Act
SEP	Salmon Enhancement Program
SPA	Scale Pattern Analysis
S/R	Stock/Recruitment
SSC	Yukon Salmon Sub-Committee
TAC	Total Allowable Catch
UFA	*Umbrella Final Agreement* of the Yukon First Nations Land Claims
USF&WS	United States Fish and Wildlife Service
WSP	*Wild Salmon Policy*
YRP	*Yukon River Panel*
YRSA	*Yukon River Salmon Agreement*
YSCCC	Yukon Salmon Conservation Catch Card